

ENCYCLOPEDIA OF MATHEMATICS AND ITS APPLICATIONS

VOLTERRA INTEGRAL AND FUNCTIONAL EQUATIONS

G.GRIPENBERG, S-O.LONDEN
& O.STAFFANS



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ENCYCLOPEDIA OF MATHEMATICS AND ITS APPLICATIONS

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Volterra Integral and Functional Equations

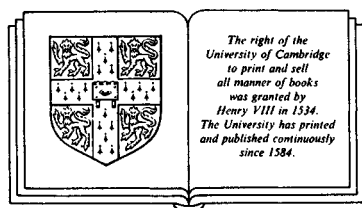
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Volterra Integral and Functional Equations

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PREFACE

During the past 25 years the theories of Volterra integral equations, Volterra integrodifferential equations, and functional differential equations have undergone rapid developments. What began as a few scattered papers on specific equations, and on particular applied problems, has grown to branches of applied analysis of considerable size, having rich structures of their own. The growth has been strongly promoted by the large number of applications that these theories have found in physics, engineering, and biology. Our understanding of those comparatively simple, usually linear, problems that were present from the beginning has increased significantly. In addition, knowledge has been obtained about more general and about more complex equations. We observe, for example, that the main part of the asymptotic theory for nonlinear Volterra integral and integrodifferential equations is of fairly recent origin. The same observation can be made about equations involving parameters or additional variables, and more generally about equations in infinite-dimensional spaces.

It is very common to make a fairly sharp distinction between Volterra integral and integrodifferential equations on the one hand, and functional differential equations on the other hand. In many respects this distinction is artificial, being due more to the different backgrounds of the researchers than to any inherent differences. In the sequel we refer to these groups of equations as 'Volterra integral and functional equations', or 'Volterra equations' for short, and many of the results that we give apply equally well to functional differential equations and to Volterra integral or integrodifferential equations. However, our general attitude to these equations resembles the attitude that has been prevailing for the last few years among people working in Volterra integral and integrodifferential equations. In particular, we pay no attention to questions that are specific for finite delay problems.

Although our knowledge of Volterra equations is swiftly growing, the point may well be made that the study of Volterra equations in finite-dimensional spaces has reached a certain maturity, which both motivates and makes possible a coherent presentation. This book testifies to the fact that we share this opinion. Certainly, we do not believe that Volterra equations in $\mathbf{R}^n$ or $\mathbf{C}^n$ constitute a closed chapter. It suffices to consider the scarcity of results on periodic solutions of nonlinear equations, or the so far unsatisfactory resolvent theory for nonintegrable kernels, to realize that the field is wide open for progress. But we do believe that the available research results make up a certain whole which may profitably be presented in the

form of a book. In addition, this will make articles scattered throughout the literature readily available to the working mathematician.

The class of ordinary differential equations is subsumed under the class of Volterra equations. However, the importance of this fact should not be exaggerated. The present text makes it clear that the theory of Volterra equations exhibits a rich variety of features not present in the theory of ordinary differential equations. The converse statement also holds: some of the more specific results available for ordinary differential equations cannot possibly be extended to Volterra equations.

Chapter 1 gives an introduction to Volterra equations and a general overview of the contents. The remaining part of the book comprises 19 chapters and is divided into three parts. In Part I (Chapters 2–10) we consider the linear theory, and in Part II (Chapters 11–15) we deal with quasilinear equations and existence problems for general nonlinear equations, and give some general asymptotic results. Part III (Chapters 16–20) is devoted to frequency domain methods in the study of nonlinear equations.

Obviously, to some extent the contents and the general approach to the subject reflect the research interests of the authors. This is particularly true of Chapters 15–20. In fact, a presentation of the results in Part III provided the initial impetus for the writing of this book. It was soon realized, however, that a treatment of the linear theory, together with the quasilinear and existence chapters of Part II, was equally important. The project thus grew into an exposé of both linear and nonlinear Volterra equations in finite-dimensional spaces. Within these limits, we have attempted to incorporate as much as possible of what we feel is essential to the understanding of Volterra equations. Where limitation of space has prohibited the inclusion of the complete proof, we have tried to give at least an outline. Additional results are frequently touched upon in the comment sections at the end of each chapter.

The major part of the text is made up of results that have so far at most been available in research papers. In particular, this is true for Chapters 4–10, 12, and 15–20. A number of the results given here have not appeared in print before, and many previously existing results have been modified and improved.

At an early stage it was decided not to include a number of worthy topics in Volterra equations: equations in infinite-dimensional spaces, stochastic equations, geometric and degree theory, attractors, stable and unstable manifolds, eigenfunction expansions, bifurcation theory, control and optimization problems, oscillation results, and numerical methods. Very little is said about neutral equations. The primary motivation for these omissions is simply a question of volume. There is more than enough in the basic theory of Volterra equations in finite-dimensional spaces to fill a good-sized book. A particular motivation was that many of these fields, for example abstract Volterra equations, are still in a state of flux, with new basic

results continuously emerging.

With a few exceptions, the entire text analyses n -dimensional rather than scalar equations. This, of course, gives the results a greater generality and a wider applicability, and it facilitates generalizations to infinite-dimensional spaces. In Parts I and II this setting is quite natural, and it does not overly complicate the proofs. The n -dimensional results in Chapters 16–19 have appeared earlier in the literature almost exclusively in a scalar setting. It does, however, appear natural and consistent to present the nonlinear asymptotic theory in the same general n -dimensional setting as we use in the earlier chapters. The theory of scalar kernels of positive type can be extended to matrix-valued kernels without too much additional complexity, and this is done in Chapter 16. In some of the theorems in Chapters 17–19 one has to make the significant additional assumption that the nonlinearity is the gradient of a scalar-valued function; when $n = 1$ this assumption is automatically satisfied.

A substantial part of the text concerns equations with measure-valued kernels. In particular, this is true for Chapters 3, 4, 10, 16, and for parts of Chapters 17–19. Our results may therefore be applied to delay and differential-delay equations. The approach that we take to functional differential equations differs, at least in appearance, from the common one. The main difference is that we stress resolvent theory, and avoid semigroup techniques. In Chapter 8 a semigroup theory is presented briefly, but the topic is not pursued further. One consequence of our approach is that there is no need for a separate treatment of functional differential equations with finite delay. Hence, we do not discuss typical finite delay questions, such as eigenfunction expansions, solutions that vanish after a finite time, and backward continuation of solutions. Instead, we attach primary importance to the integrability properties of the kernels and the resolvents. Also, we put the emphasis on a study of the Laplace transform of the kernel instead of on a study of the spectrum of the generator of the semigroup (which, however, in reality amounts to the same thing).

Every effort has been made to achieve a reasonably self-contained presentation. We have been forced to assume some basic knowledge of analysis, and do occasionally fall back on results in Rudin [2] and [3], Hille and Phillips [1], and Hewitt and Stromberg [1]. Even so, a significant part of the text has been devoted to analysis results which are difficult to find in the literature, but which are needed for the development of the theory. In particular, we present a number of theorems on vector measures, on convolutions, and on Laplace transforms. These results have been separated from the main body of the text, and several chapters contain appendices describing results of a technical nature.

Exercises and some comments on further references, historical developments, etc., are included at the end of each chapter. We warn the reader that the exercises cover a wide range of difficulty. Some of the problems are easy applications of the text, others require a much deeper understanding

of the topic.

One of our intentions when writing this book was to stimulate further research. Consequently, we have tried to make the text reflect as much as possible of the present state of knowledge in the area of the theory of Volterra equations that we cover. In the case where a result included in the text is not the best available, we have tried to incorporate a statement to this effect, together with further references in the corresponding comment section.

We sustain a hope that the text will bring about more interaction between research workers in the two fields of functional differential equations on the one hand and Volterra integral and integrodifferential equations on the other hand. We believe that both areas, which have been unduly separated from each other, would profit from such an interaction. In addition, we hope that an even larger number of concrete applications would find their way into the present theory of Volterra equations. Without the feedback from and the incitement supplied by such applications in physics, engineering, and biology, the future of the field would look much bleaker.

This text was composed and typeset at the Institute of Mathematics of the Helsinki University of Technology. It is a pleasure for us to acknowledge the excellent working conditions and facilities at our Institute that made this voluminous venture possible. At various stages of our efforts the Academy of Finland relieved some of us from teaching and administrative duties. Needless to say, we appreciate this support. Visits to Madison, Blacksburg, Graz, and Carnegie-Mellon also contributed to the final product.

Over the years John A. Nohel has actively encouraged us in our endeavour. We express our gratitude for his continuous and enthusiastic support. Our colleagues at Virginia Tech, Kenneth B. Hannsgen and Robert L. Wheeler, have given us much time and advice. In particular we wish to thank them for their fruitful criticism of earlier versions of the manuscript. Our thanks also go to Ismo Sedig for comments on Chapter 15.

We are very pleased to see our work appear in the series 'Encyclopedia of Mathematics' of the Cambridge University Press and the pleasant cooperation with David Tranah is gratefully acknowledged.

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Gustaf Gripenberg
Stig-Olof Londen
Olof Staffans

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LIST OF SYMBOLS

Basic Sets and Spaces

$\mathbf{R}$	$(-\infty, \infty)$.
$\mathbf{R}^+$	$[0, \infty)$.
$\mathbf{R}^-$	$(-\infty, 0]$.
$\mathbf{C}$	The complex plane.
$\mathbf{E}$	Either $\mathbf{R}$ or $\mathbf{C}$.
$\mathbf{Z}$	The set of integers.
$\mathbf{N}$	The set of nonnegative integers.
$\mathbf{R}^n$	The set of n -dimensional column vectors with real entries; various norms and inner products.
$\mathbf{C}^n$	The set of n -dimensional column vectors with complex entries; various norms and inner products.
$\mathbf{R}^{n \times m}$	The set of $(n \times m)$ -dimensional matrices with real entries; a norm adapted to the corresponding norms in $\mathbf{R}^m$ and $\mathbf{R}^n$.
$\mathbf{C}^{m \times m}$	The set of $(n \times m)$ -dimensional matrices with complex entries; a norm adapted to the corresponding norms in $\mathbf{C}^m$ and $\mathbf{C}^n$.
$\mathcal{T}_S$	The real line $\mathbf{R}$ where the points $t+mS$, $m \in \mathbf{Z}$, are identified.
$\emptyset$	The empty set.

Function Spaces

$V(J; Q)$	Functions of type V ($= L^p, BC$ etc.) with domain J and range contained in Q ($= \mathbf{R}^n, \mathbf{C}^{m \times n}$ etc.).
$V(J; \eta; Q)$	Functions of type V ($= L^p, BC$ etc.) with domain J , weight function η , and range contained in Q .
$V_{\text{loc}}(J; Q)$	Functions of type V with domain J that belong to $V(K; Q)$ for every compact subset K of J . If $V(K; Q)$ is a Banach space, then $V_{\text{loc}}(J; Q)$ is given the topology induced by the

	metric $d(\psi, \phi) = \sum_{j=1}^{\infty} \{2^{-j} \ \psi - \phi\ _{V(K_j)} / (\ \psi - \phi\ _{V(K_j)} + 1)\}$ where $\bigcup_{j=1}^{\infty} K_j = J$.
$V(\mathcal{T}_S; Q)$	S -periodic functions whose restrictions to each finite interval are of type V with range in Q ; $\ \phi\ _{V(\mathcal{T}_S)} = \ \phi\ _{V([0, S])}$.
$U \supset V$	The space of functions of the type $\{f + g \mid f \in U, g \in V\}$, where U and V are some function spaces with $U \cap V = \{0\}$.
AC	Absolutely continuous functions.
AP	Almost-periodic functions; sup-norm.
B^{∞}	Borel measurable, bounded functions; sup-norm.
B_0^{∞}	Functions in B^{∞} that tend to zero at infinity.
B_{ℓ}^{∞}	Functions in B^{∞} with limits at infinity (not necessarily the same limit at $-\infty$ and $+\infty$ if the domain is $\mathbf{R}$).
B_{loc}^{∞}	Functions locally in B^{∞} , i.e., their restrictions to compact subsets of the domain belong to B^{∞} ; see $V_{\text{loc}}(J, Q)$ above.
$BBV(J)$	Functions in $BV_{\text{loc}}(J)$, where $J = \mathbf{R}$ or $\mathbf{R}^+$, satisfying $\ \phi\ _{BBV(J)} = \sup_{[s, s+1] \subset J} \ \phi\ _{BV([s, s+1])} < \infty$.
$BBV_0(J)$	Functions in $BBV(J)$ such that $\ \phi\ _{BV([s, s+1])}$ tends to zero at infinity. Here $J = \mathbf{R}$ or $\mathbf{R}^+$.
BC	Bounded continuous functions; sup-norm.
BC_0	Functions in BC that tend to zero at infinity.
BC_{ℓ}	Functions in BC with limits at infinity (not necessarily the same limit at $-\infty$ and $+\infty$ if the domain is $\mathbf{R}$).
$BL^p(J)$, $1 \leq p \leq \infty$:	Functions in $L_{\text{loc}}^p(J)$ with a finite norm $\ \phi\ _{BL^p(J)} = \sup_{(s, s+1] \subset J} \ \phi\ _{L^p(s, s+1)} < \infty$. Here $J = \mathbf{R}$ or $\mathbf{R}^+$.
$BL_0^p(J)$, $1 \leq p \leq \infty$:	Functions in $BL^p(J)$ such that $\ \phi\ _{L^p(s, s+1)}$ tends to zero at infinity. Here $J = \mathbf{R}$ or $\mathbf{R}^+$.
BUC	Bounded uniformly continuous functions; sup-norm.
BUC^1	Bounded uniformly continuous functions with bounded uniformly continuous derivative.
BV	Functions of bounded variation; the norm is the sum of the total variation and the sup-norm.
BV_{loc}	Functions locally of bounded variation, i.e., their restrictions to compact subsets of the domain are of bounded variation, see $V_{\text{loc}}(J, Q)$ above.
C	Continuous functions; sup-norm on each compact set. The same space as BC_{loc} .
C_c	Continuous functions with compact support.
C_c^{∞}	Infinitely many times differentiable functions with compact support.

L^p , $1 \leq p < \infty$:	Measurable functions with finite norm $\{\int \phi(t) ^p dt\}^{1/p}$.
L^∞	Measurable functions with finite norm $\text{ess sup} \phi(t) $.
L_0^∞	Functions in L^∞ with an essential limit 0 at infinity.
L_ℓ^∞	Functions in L^∞ with essential limits at infinity (not necessarily the same limit at $-\infty$ and $+\infty$ if the domain is $\mathbf{R}$).
L_{loc}^p , $1 \leq p \leq \infty$:	Functions locally in L^p , i.e., their restrictions to compact subsets of the domain belong to L^p ; see $V_{\text{loc}}(J, Q)$ above.
$W^{1,p}$, $1 \leq p \leq \infty$:	Locally absolutely continuous functions that together with their derivative belong to L^p ; norm $\ \phi\ _{W^{1,p}(J)} = \ \phi\ _{L^p(J)} + \ \phi'\ _{L^p(J)}$.
$\tilde{L}^1$, $\hat{L}^1$, $\widehat{M}$	Functions that are Fourier or Laplace transforms of L^1 -functions or finite measures; see Sections 6.3 and 7.3.
$\mathcal{D}$	Infinitely many times differentiable functions with compact support.
$\mathcal{D}'$	Distributions; the dual of $\mathcal{D}$.
$\mathcal{S}$	Infinitely many times differentiable functions that decay rapidly at infinity.
$\mathcal{S}'$	Tempered distributions; the dual of $\mathcal{S}$.
$\mathcal{V}$	Volterra kernels of various types; see Chapters 9 and 10.
$\mathcal{F}$	Fredholm kernels of various types; see Chapter 9.
$\mathcal{K}$	This letter stands for one of the letters $\mathcal{F}$ or $\mathcal{V}$.
$[V; J]$	Kernels of type V on J .

Measure Spaces

$M(J; Q)$	Finite measures on J with range contained in Q and total variation norm; see Section 3.2.
$M(J; \eta; Q)$	Measures on J with range contained in Q and total variation norm weighted by the function η ; see Section 4.3.
$M_{\text{loc}}(J; Q)$	Set functions on J that belong to $M(K; Q)$ for every compact subset K of J .
PT	Kernels of positive type; see Chapter 16.
PT_{aco}	Kernels of anti-coercive type; see Section 16.5.
$\mathcal{M}$	Nonconvolution measure kernels of various types; see Chapter 10.

Transforms

$\hat{\mu}$	Laplace–Stieltjes transform of the measure μ , $\hat{\mu}(z) = \int_{\mathbf{R}^+} e^{-zs} \mu(ds)$ if μ is measure on $\mathbf{R}^+$, and $\hat{\mu}(z) = \int_{\mathbf{R}} e^{-zs} \mu(ds)$ if μ is a measure on $\mathbf{R}$.
$\tilde{\mu}$	Fourier–Stieltjes transform of the measure μ , $\tilde{\mu}(\omega) = \hat{\mu}(i\omega)$.
$\hat{a}$	Laplace transform of the function a , $\hat{a}(z) = \int_0^\infty e^{-zs} a(s) ds$ if a is defined on $\mathbf{R}^+$, and $\hat{a}(z) = \int_{-\infty}^\infty e^{-zs} a(s) ds$ if a is defined on $\mathbf{R}$.
$\tilde{a}$	Fourier transform of the function a , $\tilde{a}(\omega) = \hat{a}(i\omega)$.

Operators and various other Symbols

$\langle u, v \rangle$	Inner product of u and v .
$ v $	Absolute value of v if v is a scalar; norm of v if v is a vector or a matrix.
$ \mu (E)$	Total variation measure, $\sup \sum_{j=1}^N \mu(E_j) $, where $\{E_j\}_{j=1}^N$ is a partition of E .
$ v _+$	$\max\{0, v\}$. Here v is real.
$ v _-$	$\max\{0, -v\}$. Here v is real.
$\ \phi\ _V$	Norm of ϕ in the space V . Here ϕ is a function or a measure.
$\ \phi\ _{\sup(J)}$	Supremum of $ \phi $ on J .
$\ \phi\ _{\text{var}(J)}$	Total variation of ϕ on J .
$\ k\ _{L^p(J)}$	$\sup_{\ f\ _{L^q(J)} \leq 1} \int_J \int_J g(t)k(t, s)f(s) ds dt$, $1/p + 1/q = 1$; see Section 9.2.
$\ k\ _{L^{p,q}(J)}$	$\sup_{\ g\ _{L^{q'}(J)} \leq 1} \int_J \int_J g(t)k(t, s)f(s) ds dt$, $1/q + 1/q' = 1$; see Section 9.2.
$\ k\ _{B^\infty(J)}$	$\sup_{t \in J} \int_J k(t, s) ds$; see Section 9.5.
$\ \kappa\ _{B^\infty(J)}$	$\sup_{t \in J} \kappa (t, J)$; see Section 10.2.
$\ \kappa\ _{[B^\infty; L^1](J)}$	$\int_J \kappa (t, J) dt$; see Section 10.3.
$\ k\ _{[M; B^\infty](J)}$	$\sup_{t, s \in J} k(t, s) $; see Section 10.3.
$\Re$	The real part, or the symmetric part of a matrix.
$\Im$	The imaginary part, or the anti-symmetric part of a matrix.
$\sigma(\varphi)$	The spectrum of the function φ ; see Section 15.4.
$\Gamma(\varphi)$	The limit set of the function φ ; see Section 15.2.
τ_h	Translation operator: $(\tau_h \varphi)(t) = \varphi(t + h)$.

$*$	Convolution (see Sections 2.2, 3.2 and 4.1).
$\star$	The convolution-like product defined in Sections 9.2, 10.2 and 10.3.
$\succeq$	$A \succeq 0$ if $\langle y, Ay \rangle \geq 0$ for all $y \in \mathbf{C}^n$, where $\langle \cdot, \cdot \rangle$ is some given inner product; $u \succeq v$ if $u - v \in K$ where K is a given cone.
$\succ$	$A \succ 0$ if $\langle y, Ay \rangle > 0$ for all $y \in \mathbf{C}^n \setminus \{0\}$; $u \succ v$ if $u - v$ belongs to the interior of a given cone K .
$\bar{z}$	Complex conjugate of z .
$\overline{\Omega}$	Closure of the set Ω .
$\alpha\{\Omega\}$	In Chapter 12, the measure of noncompactness of the set Ω .
$\text{conv}\{\Omega\}$	Convex hull of the set Ω .
$\text{diam}\{\Omega\}$	Diameter of the set Ω ; $\text{diam}\{\Omega\} = \sup\{\ x - y\ \mid x, y \in \Omega\}$.
A^*	The adjoint of the matrix A with respect to some given inner product.
$\det[A]$	Determinant of the matrix A .
$\text{diag}[\alpha_1, \dots, \alpha_n]$	Diagonal matrix with entries $\alpha_1, \dots, \alpha_n$.
$\ker[A]$	Kernel (nullspace) of the matrix A .
$\text{range}[A]$	Range of the matrix A .
$\text{span}\{\Omega\}$	Linear span of the set Ω of vectors.
$\ln$	Logarithm with base e .
$\lim$	Limit.
$\limsup$	Superior limit.
$\liminf$	Inferior limit.
$\inf$	Infimum.
$\sup$	Supremum.
ess inf	Essential infimum.
ess lim	Essential limit.
ess sup	Essential supremum.
$[\nu, \phi]$	The value of the distribution ν evaluated at the test function ϕ .
$[\mu, \phi]$	$\Re \int_{\mathbf{R}} \langle \phi(t), (\mu * \phi)(t) \rangle dt$; see Section 16.2.
δ_t	Unit point mass at t , $\delta = \delta_0$.
μ_a	Absolutely continuous part of the measure μ .
μ_d	Discrete part of the measure μ .
μ_s	Singular part of the measure μ .
χ_E	Characteristic function of the set E .
I	Identity matrix.

$Q(\phi, \mu, T)$	$\int_0^T \langle \phi(t), \int_{[0,t]} \mu(ds) \phi(t-s) \rangle dt$, used in Chapters 17–19 .
$T_{\max}$	In Chapter 12 , the upper limit for the interval of existence of a noncontinuable solution.
x_M	In Section 13.4 , maximal solution.
x_m	In Section 13.4 , minimal solution.

1

Introduction and Overview

We give a brief introduction to Volterra equations, review some simple mathematical models involving such equations, and give an overview of the contents of the book.

1. Introduction

Let $T(\xi, t)$ be the temperature in a semi-infinite ($\xi \geq 0$) one-dimensional bar. Assume that the bar loses energy at a rate proportional to $T(0, t)$ at the point $\xi = 0$, and that an external source generates heat proportional to the function $q(t)$ at this end of the bar. If the bar is insulated at all other parts, and has temperature zero at time zero, then T is a solution of the following initial-boundary value problem (where some proportionality constants have been set equal to 1):

$$\left. \begin{aligned} T_t(\xi, t) &= T_{\xi\xi}(\xi, t), & \xi > 0, \quad t > 0, \\ T_\xi(0, t) &= \alpha T(0, t) - q(t), & t > 0, \\ T(\xi, 0) &= 0, & \xi \geq 0, \\ \lim_{\xi \rightarrow \infty} T(\xi, t) &= 0, & t \geq 0. \end{aligned} \right\} \quad (1.1)$$

Using Laplace transforms it is not hard to show that T satisfies the equation

$$T(\xi, t) = \frac{1}{\sqrt{\pi}} \int_0^t (t-s)^{-\frac{1}{2}} e^{-\xi^2/(4(t-s))} (q(s) - \alpha T(0, s)) ds, \quad t > 0, \quad \xi \geq 0.$$

Consequently, the function T can be obtained once $T(0, t)$ is known. Taking $\xi = 0$ above, we get the convolution Volterra integral equation

$$x(t) + \int_0^t k(t-s)x(s) ds = f(t), \quad t \geq 0, \quad (1.2)$$

where $x(t) = T(0, t)$, $k(t) = \frac{\alpha}{\sqrt{\pi}} t^{-\frac{1}{2}}$, and $f(t) = \frac{1}{\sqrt{\pi}} \int_0^t (t-s)^{-\frac{1}{2}} q(s) ds$.

Equation (1.2) is a typical example of the equations to be studied in this book. Of course, one might just as well write the equation in the form

$$x(t) = \int_0^t h(t-s)x(s) ds + f(t), \quad t \geq 0,$$

where $h(t) = -k(t)$, but we shall usually write the integral on the left-hand side. (Observe that, in this example, the forcing function f is of the special form $\int_0^t k(t-s)\phi(s) ds$. In some of our results we shall take advantage of this fact.) In (1.2), the unknown function $x(t)$ appears both by itself and under the integral sign multiplied by the kernel k . This fact qualifies this equation as a Volterra equation of the second kind.

As we will see later, the existence of a solution of an equation of the second kind usually presents less of a problem. A more difficult and challenging task is to determine how the solution behaves as $t \rightarrow \infty$.

As another example, let us briefly consider a control system with negative, delayed feedback. The simplest model for such a system is the differential-delay equation

$$\begin{aligned} x'(t) &= -ax(t-\tau), \quad t \geq 0; \\ x(t) &= \varphi(t), \quad t \in [-\tau, 0], \end{aligned} \tag{1.3}$$

where $\tau > 0$. One possible approach to solve this equation is to use semi-group theory, but in this book we will mostly take this equation as an example of the integrodifferential equation

$$x'(t) + \int_{[0,t]} \mu(ds)x(t-s) = f(t), \quad t \geq 0; \quad x(0) = x_0. \tag{1.4}$$

To get (1.3) from (1.4), let μ be a point mass with weight a at the point τ , i.e., $\mu(E) = a$ if $\tau \in E$; otherwise $\mu(E) = 0$. Furthermore, define f by $f(t) = -a\chi_{[0,\tau)}(t)\varphi(t-\tau)$, and let $x_0 = \varphi(0)$.

A common way of writing a linear functional differential-delay equation is

$$\begin{aligned} x'(t) &= \int_{(-\infty,0]} \eta(ds)x(t+s) + F(t), \quad t \geq 0; \\ x(t) &= \varphi(t), \quad t \leq 0, \end{aligned}$$

where the initial function φ is prescribed. However, by defining $f(t) = F(t) + \int_{(-\infty,-t)} \eta(ds)\varphi(t+s)$ and writing $\mu(E) = -\eta(-E)$, this equation may as well be reduced to (1.4).

We shall frequently use the fact that (1.4) can be put into the form (1.2) by integration.

The two preceding examples share one property: neither of them can be reduced to a finite-dimensional ordinary differential equation. Another common feature is that, in both cases, the feedback is time invariant, i.e., the influence of the past values $x(s)$ on $x(t)$ (or on $x'(t)$) depends only on the difference $t-s$, and not on the specific values of t and s . In some

cases this assumption is too restrictive; more realistic models often call for time-dependent feedback. The equation (1.2) should then be replaced by

$$x(t) + \int_0^t k(t, s)x(s) \, ds = f(t), \quad t \geq 0. \quad (1.5)$$

A similar change is required in (1.4). Equation (1.5) is a nonconvolution Volterra equation of the second kind.

In the equations above the feedback is assumed to be linear; normally this is an acceptable first approximation. However, in a more detailed analysis one must often take nonlinearities into account. For example, the state at time t may depend on $x(s)$ for $s < t$ through a nonlinear function g (and through the feedback kernel k). If both the nonlinearity and the feedback are taken to be time invariant, then we end up with the nonlinear integral equation

$$x(t) + \int_0^t k(t-s)g(x(s)) \, ds = f(t), \quad t \geq 0,$$

as the description of our system. To complicate things further, we may have a nonlinear, time-dependent feedback, we may have time-dependent nonlinearities, etc. Note that one need not always have the nonlinearity inside the integral. If, for example, one takes $y(t) = g(x(t))$ above, then y satisfies the equation

$$y(t) = g\left(f(t) - \int_0^t k(t-s)y(s) \, ds\right), \quad t \geq 0.$$

A feature common to all of the integral or integrodifferential equations above is the fact that the supports of the kernels k and μ are such that the independent variable t appears as the upper limit of integration. This is, of course, only another way of stating that the present depends on the past, but not on the future.

Equations involving integrals where the integration does not extend past the independent variable take their name from Vito Volterra, who was one of the first to investigate equations of this type. (For a short biography of Volterra, 1860–1940, see Volterra [4]. This reference also contains a complete list of Volterra's publications.) The class of equations which one gets by dropping this requirement on the upper limit of integration, i.e., equations of the form

$$x(t) + \int_0^b k(t, s)x(s) \, ds = f(t), \quad 0 \leq t \leq b,$$

are called integral equations of Fredholm type. Thus, formally a Volterra equation is a special case of a Fredholm equation. However, much of the classical theory of Fredholm equations reduces to mere trivialities when applied to Volterra equations. On the other hand, Volterra equations exhibit a variety of phenomena unknown to Fredholm theory. In addition, it is generally true that Fredholm equations describe boundary value problems and are related to operator theory, whereas Volterra equations describe

initial value problems and are related to dynamical systems. Hence there is every reason to consider these fields separately; the fact that integrals appear in both types of equations should not make one overlook the basic differences. This book concentrates on the Volterra theory, but those results that are common to Volterra and Fredholm equations are formulated in such generality that they can be applied to Fredholm equations as well. This is, in particular, true for equations that are required to hold on the whole real line $(-\infty, \infty)$.

The overwhelming majority of the integral equations we analyse are of the second kind. Only in Chapter 5 do we treat equations of the first kind in more detail. These are equations where the unknown function only appears under the integral sign. Thus,

$$\int_0^t k(t-s)x(s) ds = f(t), \quad t > 0,$$

is a convolution Volterra equation of the first kind. Analogously,

$$\int_0^t k(t,s)x(s) ds = f(t), \quad t > 0,$$

is a nonconvolution Volterra equation of the first kind. Trivially, if the kernel k and the nonhomogeneous term f are sufficiently smooth and k satisfies $0 < |k(0)| < \infty$, then the former equation may be reduced to an equation of the second kind by differentiation. A similar comment applies to the nonconvolution equation.

2. Some Examples of Volterra Equations

Volterra equations appear in a variety of applied problems. Below we give a few examples.

2.1 Example

A model for cosmic ray transport was presented in Klimas and Sandri [1] and subjected to further investigation in Hanson *et al.* [1] and [2]. It is shown that the transport of charged particles in a turbulent plasma, e.g., the cosmic rays in the interplanetary solar wind, is described by the coupled equations

$$\begin{aligned} \frac{\partial I}{\partial \tau} + \alpha \nabla \cdot \Phi &= 0, \\ \frac{\partial \Phi}{\partial \tau} + \epsilon \Omega \cdot \Phi &= -(\epsilon \eta)^2 \int_0^\tau \mathcal{K}(\epsilon, \lambda) \cdot \Phi(\tau - \lambda) d\lambda - \frac{1}{3} \alpha \nabla I. \end{aligned} \quad (2.1)$$

Here I is the omnidirectional intensity, Φ is the flux, and the parameters $(\epsilon \eta)^2$ and α are small. The term $\frac{1}{3} \alpha \nabla I$ is a source term due to gradients in the density. The tensor kernel $\mathcal{K}(\epsilon, \lambda)$ is computed from the power spectrum

associated with the random magnetic field, and decays as $\frac{1}{\lambda}$ for λ large. The fact that the kernel is nonintegrable over $\mathbf{R}^+$ causes the adiabatic approximation to be invalid. (See Klimas and Sandri [1] for details.)

For (2.1), the scalar approximation

$$\frac{\partial}{\partial t} F(t, \alpha) + \alpha \int_0^t k(t-s) F(s, \alpha) ds = -\sigma(t), \quad t > 0; \quad F(0, \alpha) = 1, \quad (2.2)$$

was presented in Hanson *et al.* [2]. The kernel $k(t)$ is assumed to satisfy $k(t) = (1+t)^{-1}$; α plays the role of $(\epsilon\eta)^2$ in (2.1). The nonhomogeneous term $\sigma(t)$ represents the density gradient. The goal is to acquire information on the behaviour of the flux $F(t, \alpha)$, in particular as $t \rightarrow \infty$ and $\alpha \downarrow 0$.

In Hanson *et al.* [1], a detailed analysis of (2.2) is carried out using Laplace transforms. Several results on how $F(t, \alpha)$ behaves for small α are obtained. See also Hanson *et al.* [2] for numerical work, and Ling [1] for some qualitative extensions. Here we observe that, by Theorem 3.3.3, one has $F(t, \alpha) = r_\alpha(t) - (r_\alpha * \sigma)(t)$ where r_α is the differential resolvent of $\alpha/(1+t)$. Since the kernel is completely monotone, one may apply Theorem 5.4.1. This leads to the conclusion that for every $\alpha > 0$ the function r_α is both bounded and integrable on $\mathbf{R}^+$. Consequently, if σ is either bounded or integrable on $\mathbf{R}^+$, then one has $\sup_{t \in \mathbf{R}^+} |F(t, \alpha)| < \infty$. If we, in addition, employ Exercise 6.13, and assume that σ is integrable, then we conclude that this bound is uniform with respect to α . Moreover, one can show that

$$r_\alpha(t) = (\alpha t \ln^2(t))^{-1} + O\left((t \ln^3(t))^{-1}\right),$$

as $t \rightarrow \infty$. See Wong and Wong [1], Theorem 4. Further results on how $F(t, \alpha)$ and r_α depend on α may be obtained by consulting, e.g., Hannsgen [11] and [15]. See also Jordan and Wheeler [6], p. 106.

2.2 Example

Population dynamics constitutes a major source of Volterra equations, the classical reference being Volterra [5]. Let us briefly sketch one example from this field.

Consider a population having an age distribution $y(t, a)$, $t \geq 0$, $a \geq 0$. Thus $\int_A y(t, a) da$ is the number of individuals with age in the set A at time t . Assume that the process of aging and dying is modelled by the balance law

$$\frac{\partial y(t, a)}{\partial t} + \frac{\partial y(t, a)}{\partial a} = -m(a)y(t, a). \quad (2.3)$$

The nonnegative function m denotes the age-dependent death rate. Suppose that the birth process satisfies

$$y(t, 0) = \int_{\mathbf{R}^+} y(t, a)b(a) da, \quad (2.4)$$

where b is the age-dependent fertility. Finally assume that the initial age distribution $y(0, a) = \phi(a)$ is known.

If one solves (2.3) by the method of characteristics and makes use of the initial condition, then one obtains

$$y(t, a) = \begin{cases} \phi(a - t) \exp\left(-\int_0^t m(s + a - t) ds\right), & 0 \leq t < a, \\ y(t - a, 0) \exp\left(-\int_0^a m(s) ds\right), & t \geq a. \end{cases} \quad (2.5)$$

Inserting the expression for $y(t, a)$ given by (2.5) into (2.4), we obtain the linear Volterra equation

$$y(t, 0) + \int_0^t k(t - s)y(s, 0) ds = f(t), \quad t \geq 0, \quad (2.6)$$

where

$$k(t) = -b(t) \exp\left(-\int_0^t m(s) ds\right),$$

$$f(t) = \int_{\mathbf{R}^+} \phi(s) \exp\left(-\int_0^t m(s + \sigma) d\sigma\right) b(t + s) ds.$$

Equation (2.6) is the classical renewal equation; see Feller [1].

A more realistic model is obtained if one includes density-dependence as well. Then

$$\begin{aligned} \frac{\partial y}{\partial t} + \frac{\partial y}{\partial a} &= -m(a, N(t))y(t, a), \\ y(t, 0) &= \int_{\mathbf{R}^+} y(t, a)b(a, N(t)) da, \\ y(0, a) &= \phi(a), \\ N(t) &= \int_{\mathbf{R}^+} y(t, a) da. \end{aligned}$$

This model was analysed in Gurtin and MacCamy [1] and has been extensively developed. For notes on this development, and for a comprehensive account of age-dependent population dynamics in general (including a wealth of references), see Webb [1]. A detailed account of mathematical models for physiologically structured populations can be found in Metz and Diekmann [1].

For a general treatment of age-dependent processes and renewal theory, see Chapter IV of Athreya and Ney [1].

2.3 Example

The spread of an epidemic in a population of fixed size can be mathematically described as follows (Diekmann [2]). Assume that the population consists of infected ($I(t)$), and susceptible ($S(t)$) individuals with $I(t) + S(t) = P$. Further assume that transitions from the infected state to the susceptible state cannot occur. Then $I'(t)$ is the rate at which susceptible individuals become infected. We take the infected population $I(t)$ to be structured according to the time since infection; i.e., $i(t, s) ds$ denotes

the number of individuals that were infected between $t - s$ and $t - s + ds$. Then

$$I(t) = \int_{\mathbf{R}^+} i(t, s) ds, \quad t \geq 0,$$

and $\frac{\partial i}{\partial t} + \frac{\partial i}{\partial s} = 0$. Hence,

$$\begin{aligned} i(t, s) &= i(t - s, 0), & t > s, \\ i(t, s) &= i(0, s - t), & t < s. \end{aligned}$$

Moreover,

$$\frac{dS}{dt} = -\frac{dI}{dt} = -i(t, 0), \quad t > 0.$$

Suppose that there exists a nonnegative function A such that the infectivity B (the rate at which susceptibles become infected) is given by

$$B(t) = \int_{\mathbf{R}^+} i(t, s) A(s) ds.$$

Then

$$\frac{dS}{dt} = -S(t) \int_{\mathbf{R}^+} i(t, s) A(s) ds, \quad t > 0,$$

with $S(0) = S_0 > 0$ given. Divide by $S(t)$ and integrate over $(0, t)$ to obtain

$$\ln \left(\frac{S(t)}{S_0} \right) = \int_0^t (S(t-s) - S_0) A(s) ds - f(t),$$

where

$$f(t) = \int_0^t \int_{\mathbf{R}^+} i(0, \tau) A(s + \tau) d\tau ds.$$

If we let $x(t) = -\ln(S(t)/S_0)$, then we end up with the nonlinear Volterra equation

$$x(t) = S_0 \int_0^t A(t-s) g(x(s)) ds + f(t), \quad t \geq 0, \quad (2.7)$$

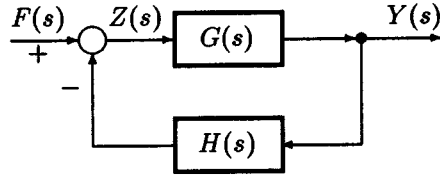
where $g(x) = 1 - e^{-x}$.

The classical paper on this model is Kermack and McKendrick [1]. An asymptotic analysis of the solutions of (2.7) was done in Diekmann [1], in Diekmann and Kaper [1], and in Gripenberg [27]. Space-dependence was incorporated in Diekmann [2]. For models including susceptible, infected, and removed individuals, see, e.g., Stech and Williams [1]. In Gripenberg [20], the spread of an infectious disease that does not induce permanent immunity is studied.

For a survey of epidemics in homogeneous populations, see Hethcote, Stech, and Van den Driessche [1]. Further work on mathematical epidemiology is to be found in Capasso, Grosso, and Paveri-Fontana [1], pp. 106–189 and in Hethcote, Lewis, and Van den Driessche [1].

2.4 Example

Volterra equations have frequently been used to describe control systems with feedback loops. We illustrate by the single closed-loop, time invariant, and linear system shown below:



Here $F(s)$ and $Y(s)$ are, respectively, the transforms of the input $f(t)$ and the output $y(t)$. The system transfer function $G(s)$ is defined as the ratio of the transform of the output $Y(s)$ to the transform of the input $Z(s)$. The feedback transfer function is denoted by $H(s)$. Consequently,

$$Y(s) = G(s) [F(s) - Y(s)H(s)],$$

or

$$Y(s) = \frac{G(s)}{1 + G(s)H(s)} F(s).$$

Let g and h be the inverse transforms of G and H , and write $k = g * h$, $f_1 = g * f$. We then have, in the time domain,

$$y(t) + \int_0^t k(t-s)y(s) ds = f_1(t), \quad t \geq 0.$$

If the feedback is built up by a nonlinear term N with no memory and a linear time invariant part with memory, then the resulting equation is

$$y(t) + \int_0^t k(t-s)N(y(s)) ds = f_1(t), \quad t \geq 0.$$

In case the system is time varying, we are led to a nonconvolution equation. Of course, a stable system is the foremost goal. In particular one wishes to have input-output stability in an L^p -sense, frequently for $p = \infty$ or $p = 2$.

The literature in the field is huge; see MacFarlane [1] for a review and reprints of important papers. For more recent work, see, e.g., Mousa, Miller, and Michel [1] and the references mentioned therein.

2.5 Example

The point model equations for a nuclear reactor with linear feedback may be written as

$$\left. \begin{aligned} \frac{d}{dt}p(t) &= \frac{k(t) - \beta}{l}p(t) + \sum_{i=1}^m \lambda_i c_i(t), \\ \frac{d}{dt}c_i(t) &= \frac{\beta_i}{l}p(t) - \lambda_i c_i(t), \quad i = 1, \dots, m, \\ k(t) &= - \int_0^t K(t-s)(p(s) - p_0) ds + f(t), \quad t \geq 0. \end{aligned} \right\} \quad (2.8)$$

Above p denotes the neutron density, k is the reactivity, β_i , $i = 1, \dots, m$, are the delayed neutron fractions ($\beta = \sum_{i=1}^m \beta_i$), l is the neutron average lifetime, and λ_i are the decay constants of the delayed neutrons. The functions c_i represent the delayed neutron densities, K is the reactivity feedback kernel, p_0 denotes the equilibrium power level, and f is the external control. It is obviously of primary interest to have stability of the equilibrium solution $p = p_0$, $c_i = \beta_i p_0 / (l \lambda_i)$.

A large number of results on stability criteria for (2.8) exist. See Londen [1], [3], and [4], Podowski [1] and [2], and the references mentioned in these articles. General treatises on nuclear reactor dynamics are, for example, Ash [1] and Akcasu, Lellouche, and Shotkin [1].

Observe that if the delayed neutrons are neglected then (2.8) reduces to

$$\frac{\left(\frac{d}{dt}p(t)\right)}{p(t)} + \frac{1}{l} \int_0^t K(t-s)(p(s) - p_0) ds = \frac{f(t)}{l}.$$

Define $x(t) = \ln(p(t)/p_0)$, $a(t) = p_0 K(t)/l$, and $h(t) = f(t)/l$. Then

$$x'(t) + \int_0^t a(t-s)(e^{x(s)} - 1) ds = h(t).$$

We end up with a nonlinear Volterra integrodifferential equation having a nonlinearity which is *a priori* bounded from below.

More complicated models may be constructed, for example, with interacting cores or including space-dependence.

2.6 Example

The theory of superfluidity has motivated several authors to study the following problem.

Let $(x, y, z) \in \mathbf{R}^3$ and let two infinite plates be located at $x = 0$ and $x = L$, respectively. Assume that the region between the plates contains liquid helium. Suppose that the boundary plates are given sinusoidal oscillations in the y -direction. This will create a one-dimensional flow in the liquid. Denote the velocity profile at the point (t, x, y, z) by $u(t, x)$. Then u will be governed by the partial differential equation

$$u_t = u_{xx}, \quad t > 0, \quad 0 < x < L,$$

subject to the conditions

$$\begin{aligned} u(0, x) &= u_0(x), \quad 0 < x < L, \\ u_x(t, 0) &= c_1(u(t, 0) - c_2 \sin(k_1 t))^3, \quad t > 0, \\ u_x(t, L) &= -c_1(u(t, L) - c_2 \sin(k_2 t))^3, \quad t > 0, \end{aligned}$$

where c_1, c_2, k_1, k_2 are positive constants, and u_0 is the initial velocity profile. This problem may be converted to the system of nonlinear Volterra

equations

$$x_1(t) + \int_0^t a_1(t-s)g_1(s, x_1(s)) ds + \int_0^t a_2(t-s)g_2(s, x_2(s)) ds = f_1(t),$$

$$x_2(t) + \int_0^t a_2(t-s)g_1(s, x_1(s)) ds + \int_0^t a_1(t-s)g_2(s, x_2(s)) ds = f_2(t),$$

where $x_1(t) = u(t, 0)$, $x_2(t) = u(t, L)$, and

$$g_i(t, u) = c_1(u - c_2 \sin(k_i t))^3, \quad i = 1, 2.$$

Clearly, these functions g_i are periodic in t . The kernels a_1 and a_2 are defined by

$$a_1(t) = \frac{1}{L} \left(1 + 2 \sum_{n=1}^{\infty} \exp(-(n\pi/L)^2 t) \right),$$

and

$$a_2(t) = \frac{1}{L} \left(1 + 2 \sum_{n=1}^{\infty} (-1)^n \exp(-(n\pi/L)^2 t) \right).$$

The nonhomogeneous terms $f_1(t)$ and $f_2(t)$ depend on the initial function u_0 . These two terms are asymptotically almost-periodic.

For this problem, one can show that solutions exist for all $t \geq 0$ and are asymptotically almost-periodic. See Miller [7], [8], and [12], and Gripenberg [21], for this result and extensions.

In Levinson [1], the corresponding semi-infinite problem is considered, i.e., the equation

$$u_t = u_{xx}, \quad x > 0, \quad t > 0,$$

subject to

$$u(x, 0) = 0, \quad x > 0; \quad u_x(0, t) = \Phi(u(0, t) - f(t)), \quad t > 0.$$

Here Φ is monotone increasing, locally Lipschitz and satisfies $\Phi(0) = 0$ and $f(t)$ is periodic. The problem is converted to

$$u(0, t) + \frac{1}{\sqrt{\pi}} \int_0^t (t-s)^{-\frac{1}{2}} \Phi(u(0, s) - f(s)) ds = 0, \quad t \geq 0,$$

and conditions are given under which $\lim_{t \rightarrow \infty} (u(0, t) - \varphi(t)) = 0$, where φ is periodic. Note that from $u(0, t)$ one may easily compute $u(x, t)$.

2.7 Example

Volterra equations are common in mathematical viscoelasticity pertaining to materials with memory. These materials are characterized by constitutive relations which are functionals of the past history of the material. In this context, we examine only a simple example and refer the reader to Renardy, Hrusa, and Nohel [1] for an in-depth overview of the field. (In particular, see Section IV 4 of this monograph.) For a brief, easily accessible account of initial value, integrodifferential equations in viscoelasticity, see Hrusa, Nohel, and Renardy [1].

Specifically, let us show how a partial integrodifferential problem can be reduced to a parameter-dependent scalar Volterra equation. For this purpose, consider

$$\left. \begin{aligned} u_{tt}(x, t) &= \sigma_x(x, t) + F(x, t), & 0 < x < 1, & & t > 0, \\ u(x, t) &= U_0(x, t), & 0 \leq x \leq 1 & & -\infty < t \leq 0, \\ u_t(x, 0+) &= U_1(x), & 0 \leq x \leq 1, & & \\ u(0, t) &= u(1, t) = 0, & & & t \geq 0, \end{aligned} \right\} \quad (2.9)$$

with the constitutive law

$$\sigma(x, t) = \int_{\mathbf{R}^+} A(s) \frac{\partial}{\partial t} u_x(x, t-s) ds.$$

Here U_0 , U_1 , F , and A are real-valued given functions. It is assumed that A is nonnegative, nonincreasing and convex on $(0, \infty)$, and integrable on $(0, 1)$. Physically, $u(x, t)$ is the displacement at time t of the point x on a one-dimensional rod located at $0 \leq x \leq 1$. The partial derivative u_x is then the strain and σ is the stress which is expressed as a convolution integral of the strain. The weight function A is the linear stress relaxation modulus and F denotes the external force. See, e.g., Hannsgen and Wheeler [5].

If the constitutive law is used to replace σ in (2.9), then one obtains, after integrating with respect to t ,

$$\begin{aligned} u_t(x, t) &= \int_0^t A(t-s) u_{xx}(x, s) ds + f(x, t), & t \geq 0, \\ u(x, 0) &= u_0(x) = U_0(x, 0), & u(0, t) = u(1, t) = 0, \end{aligned} \quad (2.10)$$

where f includes F , the initial velocity U_1 , and the strain history. Let $Lu = u''$ for $u \in \{\varphi'' \in L^2(0, 1) \mid \varphi(0) = \varphi(1) = 0\}$. Then (2.10) can be written in the abstract form

$$U'(t) = \int_0^t A(t-s) LU(s) ds + G(t), \quad t \geq 0; \quad U(0) = u_0. \quad (2.11)$$

One can show that (2.11) has the solution

$$U(t) = R(t)u_0 + \int_0^t R(t-s)G(s) ds,$$

where the resolvent R is the solution of (2.11) with $G \equiv 0$; see Carr and Hannsgen [1] and [2]. The self-adjoint operator L has the eigenvalues $\lambda_n = -(n\pi)^2$ with the corresponding normalized eigenfunctions $\varphi_n(x) = \sqrt{2} \sin(n\pi x)$. Separation of variables yields, for $y \in L^2(0, 1)$,

$$R(t)y = \sum_{n=1}^{\infty} r_n(t) \left\{ \int_0^1 y(x) \varphi_n(x) dx \right\} \varphi_n,$$

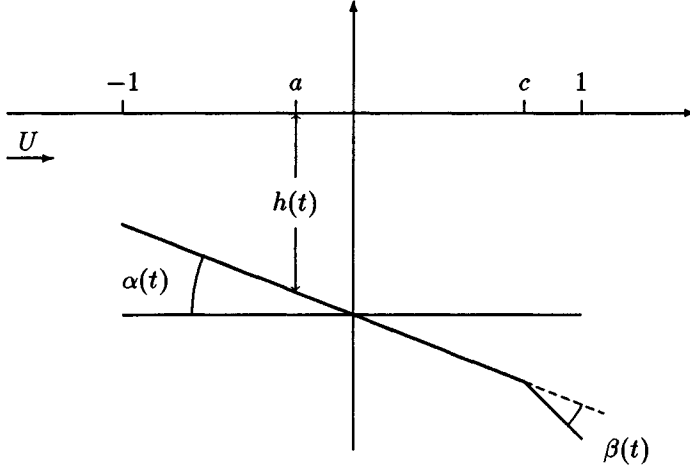
where the scalar functions r_n are the solutions of

$$r'_n(t) + |\lambda_n| \int_0^t A(t-s) r_n(s) ds = 0; \quad r_n(0) = 1.$$

Once we have the functions r_n we get R and hence also $u(x, t)$.

2.8 Example

The flutter problem for an aerofoil section in a two-dimensional flow has been considered by Burns, Herdman, and Turi [2] and Burns, Cliff, and Herdman [1]. (See these articles for a detailed derivation of the model; here we only briefly mention the structure of the resulting equations.)



Consider the aerofoil section above, and assume that the aerofoil is placed in an incompressible, inviscid two-dimensional flow with undisturbed stream velocity U . Let $u(t)$ be a control torque on the flap. Define the vector $x(\cdot) \in \mathbf{R}^8$ by

$$x(t) = (h'(t), \alpha'(t), \beta'(t), h(t), \alpha(t), \beta(t), \Gamma_q(t), \Gamma'_q(t))^T,$$

where $\Gamma_q : \mathbf{R} \mapsto \mathbf{R}$ is the total aerofoil circulation. The equations of motion may then be written

$$\begin{aligned} \frac{d}{dt} \left(A_0 x(t) + \int_{-\infty}^0 A_1(s) x(t+s) ds \right) \\ = B_0 x(t) + \int_{-\infty}^0 B_1(s) x(t+s) ds + \tilde{G} u(t), \end{aligned} \quad (2.12)$$

where A_0, A_1, B_0, B_1 and $\tilde{G}$ are certain 8×8 matrices.

The matrices A_i, B_i have a special structure; in particular, A_0 is singular. Therefore, (2.12) may be viewed as a mixture of a Volterra equation of the first kind and a Volterra equation of the second kind. See Burns, Herdman, and Turi [1] and Burns *et al.* [1] for work on this problem.

2.9 Example

Consider a one-dimensional homogeneous bar of unit length, let $u(t, x)$ denote the temperature of the bar at the point $x \in [0, 1]$ and at time $t \geq 0$, and assume that the temperature at the end-points is kept at zero.

Let the internal energy ϵ and the heat flux q be functionals of the past history of u and the gradient of u , respectively, that is, assume that the

material has memory. A reasonable mathematical model is to take

$$\begin{aligned}\epsilon(t, x) &= b_0 u(t, x) + \int_0^t \beta(t-s) u(s, x) \, ds, \\ q(t, x) &= -c_0 \sigma(u_x(t, x)) + \int_0^t \gamma(t-s) \sigma(u_x(s, x)) \, ds,\end{aligned}$$

for $t \geq 0$ and $0 \leq x \leq 1$. Here b_0, c_0 are positive constants, and $\beta, \gamma : \mathbf{R}^+ \mapsto \mathbf{R}$. The nonlinear function σ is monotone increasing, sufficiently smooth, and satisfies $\sigma(0) = 0$. Denote the externally supplied heat by $h(t, x)$. The balance equation is then

$$\epsilon_t = -\frac{\partial q}{\partial x} + h$$

which may be reformulated as

$$\begin{aligned}b_0 u(t, x) + \int_0^t \beta(t-s) u(s, x) \, ds \\ = \int_0^t c(t-s) (\sigma(u_x(s, x)))_x \, ds + G(t, x).\end{aligned}\quad (2.13)$$

The kernel c is defined by $c(t) = c_0 - \int_0^t \gamma(s) \, ds$; the function G is determined by the external heat supply and the history prior to $t = 0$.

For details and for references to the underlying physical theory, we refer the reader to Nohel [3].

Owing to the space-dependence of u , the analysis of (2.13) does *per se* fall outside the scope of the present work. Nevertheless, since the kernels β and γ are usually taken to be of positive type, the analysis does use much of the theory developed in Chapters 16–20. In particular, strong positivity and the theory of completely positive kernels are useful in this context. See Nohel [3], Sections 3.2, 4.2, and 4.3.

3. Summary of Chapters 2–20

The text is divided into three parts, devoted respectively to linear theory, general nonlinear theory, and asymptotic theory for nonlinear equations.

The first part on linear theory may further be subdivided according to the distinction convolution/nonconvolution. Chapters 2–8 deal with convolution equations and therefore rely heavily on transform theory. Chapters 9 and 10 analyse nonconvolution equations.

The general nonlinear part, Chapters 11–15, addresses questions like existence of solutions, continuous dependence, perturbations, Lyapunov methods, and general asymptotics.

In the final part, Chapters 16–20, we discuss frequency domain and monotonicity methods. Most of the equations investigated in this part

are of convolution type, and for the development of their asymptotics the theory of kernels of positive type plays a decisive role.

At the end of each chapter there are exercises and some comments on references, historical developments, etc. For a wealth of further comments on earlier work, especially on the Soviet literature, we refer the reader to Corduneanu [10].

A brief survey of the contents of each chapter follows.

PART I

3.2 Chapter 2: Linear Convolution Integral Equations

We begin our study of Volterra equations by considering the linear convolution equation

$$x(t) + \int_0^t k(t-s)x(s) \, ds = f(t), \quad t \in \mathbf{R}^+, \quad (2.1.1)$$

where k is an $n \times n$ matrix with complex entries, and x and f are complex-valued n -vectors. We use the notation $(k * x)(t) \stackrel{\text{def}}{=} \int_0^t k(t-s)x(s) \, ds$. By successive iteration, we formally derive the *variation of constants formula* $x = f - r * f$ where the *resolvent* r should satisfy the equations $r = k - k * r = k - r * k$. Before progressing further we list some basic properties of convolutions and of Laplace and Fourier transforms of functions (the proofs are deferred to an appendix). Then, in Section 2.3, we prove the existence of a unique solution r of the resolvent equations, and deduce the variation of constants formula.

Section 2.4 is devoted to the Paley–Wiener theorem. This fundamental result says that if $k \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, then $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ if and only if $(\hat{})$ denotes the Fourier transform)

$$\det[I + \hat{k}(z)] \neq 0, \quad \Re z \geq 0.$$

The proof is executed in a series of lemmas in Section 2.5. In addition to kernels supported on $\mathbf{R}^+$ (Volterra kernels) we consider kernels supported on $\mathbf{R}$ (Fredholm kernels). Thus, some of our results apply to the equation

$$x(t) + \int_{-\infty}^{\infty} k(t-s)x(s) \, ds = f(t), \quad t \in \mathbf{R}. \quad (2.4.3)$$

To realize the importance of the condition $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, observe that, if $f \in L^p(\mathbf{R}^+; \mathbf{C}^n)$ and $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, then $r * f \in L^p(\mathbf{R}^+; \mathbf{C}^n)$. Consequently, the solution $x = f - r * f$ of (2.1.1) belongs to the same space as f . Theorems 2.4.5–2.4.7 give further results along the same line.

An analysis of $f \mapsto r * f$ as a map of $L^2(\mathbf{R}^+; \mathbf{C}^n)$ into itself, together with Nyquist's theorem on the number of zeros of $\det[I + \hat{k}(z)]$ in $\Re z > 0$ conclude the chapter.

3.3 Chapter 3: Linear Integrodifferential Convolution Equations

In the third chapter we analyse the integrodifferential equations

$$x'(t) + \int_{[0,t]} \mu(ds)x(t-s) = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0, \quad (3.1.2)$$

and

$$x'(t) + \int_{\mathbf{R}} \mu(ds)x(t-s) = f(t), \quad t \in \mathbf{R}. \quad (3.1.6)$$

The kernel μ in (3.1.2) is a locally bounded matrix-valued Borel measure supported on $\mathbf{R}^+$ (a Volterra kernel), whereas μ in (3.1.6) is a bounded matrix-valued measure on $\mathbf{R}$ (a Fredholm kernel). We denote these classes of measures by $M_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ and $M(\mathbf{R}; \mathbf{C}^{n \times n})$ respectively. The functions x and f take their values in $\mathbf{C}^n$. Convolution is again denoted by $*$, that is, $(\mu * x)(t) = \int_{\mathbf{R}} \mu(ds)x(t-s)$. In case μ and x are supported on $\mathbf{R}^+$, then $(\mu * x)(t) = \int_{[0,t]} \mu(ds)x(t-s)$. Note that if μ consists of point masses only (a case which we do not exclude), then (3.1.2) reduces to a differential-delay equation.

Linear functional differential equations are included in the formulation (3.1.2), but no special consideration is given to equations with finite delay.

Following an expositional introduction, we study the convolution of a measure and a function, and consider Laplace transforms of locally finite measures. Section 3.3 contains the main results of the chapter. We first deduce the existence of the *differential resolvent* of a measure $\mu \in M_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$, that is, we prove the existence of a unique locally absolutely continuous function r defined on $\mathbf{R}^+$ satisfying $r(0) = I$ and the equations

$$r'(t) + (\mu * r)(t) = r' + (r * \mu)(t) = 0, \quad t \in \mathbf{R}^+.$$

With the aid of the differential resolvent we can express the solution of (3.1.2) in the *variation of constants* form

$$x(t) = r(t)x_0 + (r * f)(t), \quad t \in \mathbf{R}^+.$$

The second part of Section 3.3 gives the Paley–Wiener theorems applicable to (3.1.2) and (3.1.6). Assume that μ in (3.1.2) satisfies $\mu \in M(\mathbf{R}^+; \mathbf{C}^{n \times n})$. Then the resolvent r of μ satisfies $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ if and only if $\det[zI + \hat{\mu}(z)] \neq 0$ for $\Re z \geq 0$. Similarly, if μ in (3.1.6) satisfies $\mu \in M(\mathbf{R}; \mathbf{C}^{n \times n})$, then μ has a resolvent $r \in L^1(\mathbf{R}; \mathbf{C}^{n \times n})$ if and only if $\det[zI + \hat{\mu}(z)] \neq 0$ for $\Re z = 0$. If the latter resolvent (required to satisfy $r' + \mu * r = r' + r * \mu = 0$ on $\mathbf{R}$ and $r(0+) - r(0-) = I$) exists, then we can express the solution of (3.1.6) in the slightly different *variation of constants* form

$$x = r * f,$$

provided that f belongs to some suitable space of functions. Theorems 3.3.9–3.3.10 demonstrate the significance of (the existence of) integrable resolvents. We conclude the section by giving some results on periodic solutions and on L^2 -solutions.

At the end of Chapter 3 there are five appendices, i.e., Sections 3.4–3.8, containing supplementary results, needed in this as well as in later chapters. Section 3.4 formulates different versions of Fubini's theorem, and Section 3.5 discusses the total variation of a vector measure. In Section 3.6 we establish some properties of the convolution of a measure and a function. Derivatives of these convolutions are analysed in Section 3.7. Finally, in Section 3.8 we give a few results on Laplace transforms of measures.

3.4 Chapter 4: Equations in Weighted Spaces

In this chapter we analyse the convolution integral equations

$$x(t) + \int_{[0,t]} \mu(ds)x(t-s) = f(t), \quad t \in \mathbf{R}^+, \quad (4.1.1)$$

and

$$x(t) + \int_{\mathbf{R}} \mu(ds)x(t-s) = f(t), \quad t \in \mathbf{R}, \quad (4.1.2)$$

with *measure-valued* kernels. Furthermore, we give some additional results on (3.1.2) and (3.1.6). In (4.1.1) we assume that $\mu \in M_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ (a Volterra kernel), and in (4.1.2) that $\mu \in M(\mathbf{R}; \mathbf{C}^{n \times n})$ (a Fredholm kernel). If μ consists of point-masses only, then the equation becomes a difference equation.

The resolvent ρ of μ is required to satisfy $\rho + \mu * \rho = \rho + \rho * \mu = \mu$; hence ρ (if it exists) is now a measure. This fact forces us to define the convolution of two measures, and to analyse the properties of this type of convolution (the proof is deferred to an appendix). This being done, we show in Theorem 4.1.5 that a measure $\mu \in M_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ has a unique resolvent $\rho \in M_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ if and only if $\det[I + \mu(\{0\})] \neq 0$ (where $\mu(\{0\})$ denotes a possible point mass of μ at zero; observe that the integral in (4.1.1) contains a hidden term $\mu(\{0\})x(t)$, so that the actual coefficient of $x(t)$ is $I + \mu(\{0\})$). With this measure resolvent ρ at our disposal, we can write the solution of (4.1.1) in the variation of constants form $x = f - \rho * f$.

The remaining part of Chapter 4 is formulated in the framework of *weighted spaces*. Sections 4.2 and 4.3 introduce the reader to weight functions, and define weighted function spaces and the weighted measure spaces $M(\mathbf{R}^+; \varphi)$ and $M(\mathbf{R}; \varphi)$. Here the weight function $\varphi : \mathbf{R} \rightarrow \mathbf{R}^+$ is positive, locally bounded, submultiplicative (that is, $\varphi(t+s) \leq \varphi(t)\varphi(s)$), and satisfies $\varphi(0) = 1$. For example, the space $M(\mathbf{R}^+; \varphi; \mathbf{C}^{n \times n})$ consists of those matrices of locally finite Borel measures that satisfy $\int_{\mathbf{R}^+} |\mu|(ds)\varphi(s) < \infty$.

The key question of Section 4.4 is the following. Suppose that $\mu \in M(\mathbf{R}^+; \varphi; \mathbf{C}^{n \times n})$. When does it follow that the resolvent ρ belongs to the same space? A crucial role in the development of the answers (i.e., Theorems 4.4.3, 4.4.4, and their corollaries) is played by Theorem 4.4.2, due to Gel'fand, concerning the invertibility of a scalar measure. Note that in this result a size restriction must be imposed on the singular part of μ . Once we know that $\rho \in M(\mathbf{R}^+; \varphi; \mathbf{C}^{n \times n})$, or $\rho \in M(\mathbf{R}; \varphi; \mathbf{C}^{n \times n})$,

then various results about the solution (Theorems 4.4.11 and 4.4.12) can be deduced.

The integrodifferential equations (3.1.2) and (3.1.6) can be analysed in weighted spaces too. The results are actually sharper than those for the integral equation in the sense that the conditions obtained are both necessary and sufficient; see Theorems 4.4.13 and 4.4.14.

Equations of so-called neutral type are discussed among the exercises of this chapter.

3.5 Chapter 5: Completely Monotone Kernels

Scalar convolution kernels that are locally integrable and *completely monotone* on $(0, \infty)$, i.e., kernels satisfying $(-1)^j k^{(j)}(t) \geq 0$ for $t > 0$ and $j = 0, 1, 2, 3, \dots$, have been the object of numerous studies. In Chapter 5 we present some of the results obtained and apply them to Volterra equations. We extend the concept of complete monotonicity to matrices by requiring $\langle v, k(\cdot)v \rangle$ to be completely monotone for all $v \in \mathbf{C}^n$ (where $\langle \cdot, \cdot \rangle$ is some inner product).

Section 5.2 contains basic definitions of matrix algebra needed both in Chapter 5 and in Part III. The well-known theorem by Bernstein on the representation of a completely monotone function as the Laplace transform of a nonnegative measure is given. Subsequently, a characterization of the Laplace transform $F(z)$ of a completely monotone function is presented. We show that such a transform can be analytically extended to $\mathbf{C} \setminus \mathbf{R}^-$, is real on the real axis, satisfies $\Im F(z) \leq 0$ for $\Im z > 0$, and $\lim_{x \rightarrow \infty} F(x) = 0$. Conversely, a function $F(z)$ having these four properties is shown to be the transform of a completely monotone function. This result is extremely useful in applications.

As a consequence of Bernstein's theorem we establish, in Theorem 5.2.8, that the logarithm of a completely monotone function is convex.

It is well known that the resolvent of a scalar completely monotone function is completely monotone. We prove this fact for the matrix case in Theorem 5.3.1.

In Section 5.4 we analyse the differential resolvent r of a completely monotone, locally integrable function k . It is easy to show that r cannot be completely monotone on $(0, \infty)$. However, $-r$ is the sum of a completely monotone function and an exponentially decaying one. The proof of this fact contains some elaborate analysis.

Section 5.5 briefly considers linear *Volterra equations of the first kind*, i.e., equations of the type

$$\int_0^t k(t-s)x(s) ds = f(t), \quad t \in \mathbf{R}^+. \quad (5.1.1)$$

If r satisfies $(r * k)(t) = (k * r)(t) = I$ for $t > 0$, then r is a *resolvent of the first kind* of k , and the solution of (5.1.1) is $x(t) = r(t)f(0) + (r * f')(t)$. As an application of the characterization of the Laplace transform of a

completely monotone function, we obtain a result on the existence of a resolvent of the first kind. Furthermore, the case where the (scalar) kernel k is nonnegative and nonincreasing is studied in Theorem 5.5.5.

3.6 Chapter 6: Nonintegrable Kernels with Integrable Resolvents

The Paley–Wiener theorems in Chapters 2–4 which yield the existence of an integrable resolvent all require the original kernel to be integrable. In general, if the conclusion is to be retained, then we cannot remove this requirement without imposing some extra conditions. Although some answers, as to what extra conditions suffice, do exist, the problem is still to some extent open. Chapter 6 treats this question in some detail.

In Section 6.2 we show that, if in the scalar case the kernel k is locally integrable, nonnegative, nonincreasing, and convex, then $\|r\|_{L^1(\mathbb{R}^+)} < \infty$. In fact, we record an explicit upper bound for $\|r\|_{L^1(\mathbb{R}^+)}$, independent of k .

One may combine the Paley–Wiener theorems on $\mathbb{R}^+$ with this result on the size of the resolvent of a scalar convex kernel, to obtain additional results on the integrability of resolvents. To do this, it is necessary to study functions that are locally equal to Laplace transforms. This is done in Section 6.3. The actual combination is presented in Section 6.4, together with some further extensions of the Paley–Wiener results. Observe that two of the extensions, i.e., Theorems 6.4.2 and 6.4.4, relate to systems.

In the appendix in Section 6.5, we state and prove an inequality due to Hardy and Littlewood that is needed in the proof of the fact that the resolvent of a convex kernel is integrable.

3.7 Chapter 7: Unbounded and Unstable Solutions

Section 7.2 is devoted to the cases where the conditions $\det[zI + \hat{\mu}(z)] \neq 0$, $\Re z \geq 0$ or $\det[I + \hat{\mu}(z)] \neq 0$, $\Re z \geq 0$ are violated in the open right half plane $\Re z > 0$. In these cases, the differential resolvent or resolvent of μ includes exponentially growing terms. Theorems 7.2.3 and 7.2.5 describe the behaviour of solutions of equations containing such a kernel μ . We remark that these results are formulated in terms of weighted spaces.

The case where $\det[zI + \hat{\mu}(z)]$ or $\det[I + \hat{\mu}(z)]$ has zeros on the critical line is studied in Section 7.3. The results are similar to those obtained in Section 7.2, but stronger assumptions have to be made. The analysis is much more intricate, and many details are omitted. As an application, some consequences pertaining to the renewal equation are given.

3.8 Chapter 8: Volterra Equations as Semigroups

In Chapter 8 we briefly analyse Volterra equations from a *semigroup* point of view. In Section 8.2 we define the *initial function semigroup* T and the *forcing function semigroup* S , and compute their infinitesimal generators.

Theorem 8.2.8 ascertains that the adjoint semigroup of T is the transposed forcing function semigroup.

It is shown in Section 8.3 that by considering *extended semigroups* one may remove some of the limitations that appear in the context of initial function and forcing function semigroups.

This semigroup approach is not used elsewhere in the book.

3.9 Chapter 9: Linear Nonconvolution Equations

Here we present the basic theory on Volterra equations of the form

$$x(t) + \int_0^t k(t, s)x(s) ds = f(t), \quad t \in \mathbf{R}^+. \quad (9.1.1)$$

Much of the analysis is performed in the setting of various *Banach algebras*.

In Section 9.2 we first define *kernels of type L^p* on some interval J . These are the matrix-valued kernels k for which the mapping $f \mapsto \int_J |k(t, s)|f(s) ds$ takes $L^p(J)$ continuously into itself. If multiplication is defined by

$$(k_1 \star k_2)(t, s) = \int_J k_1(t, u)k_2(u, s) du,$$

then this class of kernels constitutes a Banach algebra. The subalgebra of Volterra kernels consists of those kernels for which $k(t, s) = 0$ when $s > t$. The convenient Proposition 9.2.7 gives conditions (sufficient when $1 < p < \infty$, necessary and sufficient when $p = 1$ or $p = \infty$) for k to be a kernel of type L^p .

Section 9.3 is devoted to the existence of a resolvent of a kernel of type L^p . The resolvent r of k is required to be of the same type as k , and to satisfy $r + k \star r = r + r \star k = k$ on J^2 . If k has a resolvent, then the solution of equation (9.1.1) can be written as $x = f - r \star f$.

Theorem 9.3.13 is a key result on the existence of a resolvent of a non-convolution Volterra kernel. Here it is shown that the global problem of finding a resolvent of k on J can be reduced to the local problem of finding a resolvent of k on J_i , where $J = \bigcup_{i=1}^m J_i$. It follows from Banach-algebra arguments that k has a unique resolvent on J_i if $\|k\|_{L^p(J_i)} < 1$. Corollary 9.3.18 gives the classical representation of the resolvent as a sum of iterated kernels.

In practice we often have $J = \mathbf{R}^+$. For this case, it is convenient to have local versions of the theorems giving the existence of a resolvent. Such versions are found in Section 9.4.

Kernels of so-called bounded and continuous types on J are considered in Section 9.5. Kernels of *bounded type* map $B^\infty(J)$ into itself, and kernels of *continuous type* map bounded subsets of $L^\infty(J)$ into equicontinuous sets of functions on J . Section 9.6 analyses certain subalgebras of kernels of bounded type, in particular, kernels of *bounded periodic type*, and kernels of *asymptotically periodic type*.

In the applications of fixed point theorems one frequently requires the kernel k , or the resolvent r , to generate a *compact* mapping. In Section 9.7 we formulate sufficient conditions for this to happen in the case where k is of type L^p , with $1 \leq p \leq \infty$.

For equations with nonconvolution kernels, the global theory is much more complicated than for convolution equations. Only in the scalar case has it been possible to obtain interesting results on the asymptotic behaviour of solutions of (9.1.1). These results (in Sections 9.8 and 9.9) are based on monotonicity or sign properties of the kernel k . Section 9.8 starts out by presenting a generalization of the well-known Gronwall inequality and continues with results exposing the advantage of having a nonnegative resolvent.

The appendix in Section 9.10 contains some lemmas needed earlier in the characterization of kernels of type L^p on J .

3.10 Chapter 10: Linear Nonconvolution Equations with Measure Kernels

In this chapter we investigate the equations

$$x(t) + \int_{[0,t]} \kappa(t, ds)x(s) = f(t), \quad t \in \mathbf{R}^+, \quad (10.1.5)$$

and

$$x'(t) + \int_{[0,t]} \kappa(t, ds)x(s) = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0. \quad (10.1.6)$$

The kernel κ is now assumed to be a t -dependent matrix of complex measures. Delay equations of the type

$$\begin{aligned} x(t) + a(t)x(t - g(t)) &= f(t), \\ x'(t) + a(t)x(t - g(t)) &= f(t), \end{aligned}$$

are therefore included.

We begin Section 10.2 by defining Volterra *measure kernels of type B^∞* on an interval $J \subset \mathbf{R}$. Specifically, κ is of type B^∞ on J if, for each Borel set $E \subset J$, the function $\kappa(\cdot, E)$ is Borel measurable, if κ is of Volterra type, i.e., if for each $t \in J$, $\kappa(t, \cdot)$ is supported on $J \cap (-\infty, t]$, and if $\sup_{t \in J} |\kappa|(t, J) < \infty$. The set of these kernels can be made a Banach algebra by defining the norm of κ to be $\sup_{t \in J} |\kappa|(t, J)$, and defining multiplication by $(\kappa_1 \star \kappa_2)(t, E) = \int_J \kappa_1(t, ds)\kappa_2(s, E)$.

The resolvent ρ of a kernel κ of type B^∞ is required to be of the same type as κ , and to satisfy $\rho + \kappa \star \rho = \rho + \rho \star \kappa = \kappa$. If κ has a resolvent, then the variation of constants formula $x = f - \int_J \rho(t, ds)f(s)$ for the solution of (10.1.5) is valid. Theorem 10.2.8 and Corollary 10.2.9 show that if J can be divided into subintervals J_i such that the norm of κ on each J_i is < 1 , then κ has a resolvent on J . Some local versions of these claims and a perturbation result bring the section to an end.

Section 10.3 analyses (10.1.6), and begins by defining a class of measures that generate maps of $B^\infty(J)$ into $L^1(J)$. To quote the main requirements, κ is said to be a *Volterra measure kernel of type* $[B^\infty; L^1]$ on J if the measure $\kappa(t, ds)$ is, for each $t \in J$, supported on $J \cap (-\infty, t]$, and if $\int_J |\kappa|(t, J) dt < \infty$. With $\int_J |\kappa|(t, J) dt$ as norm, these measures form a Banach space.

To supplement this space of kernels it is appropriate to define *kernels of type* $[M; B^\infty]$ mapping M (hence L^1) into B^∞ . These are simply the functions $k \in B^\infty(J^2; \mathbb{C}^{n \times n})$ that vanish for $s > t$. In Theorems 10.3.4 and 10.3.5, we prove various results on the $\star$ -product of the three different spaces of kernels that have been introduced.

The differential resolvent r of κ is required to satisfy $r(t, t) = I$, $r(t, s) = 0$, $s > t$, and

$$\begin{aligned} \frac{\partial}{\partial t} r(t, s) + \int_{[s, t]} \kappa(t, du) r(u, s) &= 0, \\ \frac{\partial}{\partial s} r(t, s) - \int_{[s, t]} r(t, u) \kappa(u, ds) du &= 0 \end{aligned}$$

(where the second relation should be interpreted as a measure equation). As Theorem 10.3.7 exhibits, if κ is locally of type $[B^\infty; L^1]$ on $\mathbf{R}^+$, then it has a unique differential resolvent r which is locally of type $[M; B^\infty]$ on $\mathbf{R}^+$. The solution of (10.1.6) can then be written as $x(t) = r(t, 0)x_0 + \int_0^t r(t, s)f(s) ds$.

Section 10.4 contains some global results for (10.1.6). It is worth pointing out that all available results of this kind are of perturbation type. In particular, they are based on the fact that if $\kappa = \kappa_1 + \kappa_2$ (both locally of type $[B^\infty; L^1]$ on $\mathbf{R}^+$), and if r_1 and r are the differential resolvents of respectively κ_1 and κ , then $r + r_1 \star \kappa_2 \star r = r + r \star \kappa_2 \star r_1 = r_1$. Various examples are given where r_1 originates in an ordinary differential equation, or in a Volterra equation of convolution type. The section is concluded with a brief discussion of equation (10.1.6) with $\mathbf{R}^+$ replaced by $\mathbf{R}$.

PART II

3.11 Chapter 11: Perturbed Linear Equations

In this chapter we launch the study of nonlinear equations by analysing perturbations of linear equations. Specifically, we consider the equations

$$x(t) + \int_0^t k(t, s)x(s) ds = G(x)(t), \quad t \in \mathbf{R}^+, \quad (11.1.1)$$

and

$$x'(t) + \int_{[0, t]} \kappa(t, ds)x(s) = G(x)(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0, \quad (11.1.2)$$

where we assume that G can be written as the sum of a small x -dependent term and a term independent of x . Using the variation of constants formula,

we rewrite equations (11.1.1) and (11.1.2) as

$$\begin{aligned}x(t) &= G(x)(t) - (r_1 \star G(x))(t), \\x(t) &= r_2(t)x_0 + (r_2 \star G(x))(t),\end{aligned}$$

respectively. Here r_1 is the resolvent of k , and r_2 is the differential resolvent of κ . The product $\star$ includes the normal convolution as a special case.

In Section 11.2, we formulate two fixed point theorems which are applied in subsequent sections. Section 11.3 treats the convolution case of (11.1.1) (i.e., $k(t, s)$ is replaced by $k(t - s)$), with $k \in L^1(\mathbf{R}^+; \mathbf{R}^{n \times n})$, $G(x)(t) = \int_0^t h(t - s, x(s)) ds$, and $h(t, y) = o(|y|)$ as $y \rightarrow 0$. Under the key assumption that $\det[I + \hat{k}(z)] \neq 0$ for $\Re z \geq 0$, we obtain asymptotic results on x , including results on the rate of convergence to zero. An analogous result is given for the corresponding equation on $\mathbf{R}$. In Section 11.4 we analyse the convolution case of (11.1.2) (then $\kappa(t, ds)x(s)$ is replaced by $\mu(ds)x(t - s)$), without giving G any more explicit form. Section 11.5 deals with perturbations of ordinary differential equations.

In the last Section 11.6, we work in the framework of $L^2(\mathbf{R}^+; \mathbf{R}^n)$. Conditions are imposed on the kernel which guarantee that the resolvent maps $L^2(\mathbf{R}^+; \mathbf{R}^n)$ into itself. Under certain hypotheses on the nonlinearities we obtain $x \in L^2(\mathbf{R}^+; \mathbf{R}^n)$. Several additional perturbation theorems are included in the exercises of Chapter 11, in particular, some theorems on nonconvolution equations.

3.12 Chapter 12: Existence of Solutions of Nonlinear Equations

In Part I, where we developed the linear theory, the existence of a unique solution followed once the existence of a unique resolvent was established. In Chapter 11, the existence of a solution is a consequence of various fixed point theorems (except for Theorems 11.3.3 and 11.4.3 where it is assumed). However, in the remaining part of the text the existence of a solution is frequently part of the hypothesis; the main point of interest being the asymptotic behaviour of the solution(s). It is therefore appropriate at this point to give some existence and uniqueness results, general enough to cover all equations analysed later.

After a simple introductory result, Theorem 12.1.1, on the existence of a continuous solution of the equation

$$x(t) = f(t) + \int_0^t a(t, s)h(s, x(s)) ds, \quad t \geq 0,$$

we state the fixed point Theorem 12.1.3 which we use for the proofs of our existence results. This theorem, based on the measure of noncompactness, is reasonably general and includes both Schauder's fixed point theorem and the contraction mapping principle.

Theorem 12.2.1 is our main result on the existence of continuous solutions. It is formulated for the quite general equation

$$x(t) = G(x)(t), \quad t \geq 0, \tag{12.1.6}$$

where G maps a suitably restricted class of continuous functions defined on $[0, T]$ into $C([0, T]; \mathbf{R}^n)$. In addition, G is required to be causal and to satisfy a certain combined contraction and equicontinuity hypothesis. The conclusion contains statements about existence and about the behaviour of maximally defined solutions. The proof of Theorem 12.2.1 uses some elaborate analysis. Corollary 12.2.2 gives an easy uniqueness result. Theorems 12.2.4 and 12.2.5 consider the question whether the solutions can be forced to stay in some *a priori* defined set. This problem is of importance for comparison studies. Section 12.2 is concluded by several existence results for

$$x(t) = f(t) + \int_0^t g(t, s, x(s)) \, ds, \quad t \geq 0, \quad (12.1.7)$$

where g satisfies the Carathéodory conditions.

Section 12.3 is devoted to the functional differential equation

$$\frac{d}{dt}x(t) = H(x)(t), \quad t \geq 0; \quad x(0) = x_0, \quad (12.3.1)$$

where the causal operator H maps a suitably restricted class of continuous functions into integrable functions. By integrating (12.3.1) and then applying Theorem 12.2.1 we obtain an existence theorem for (12.3.1).

We begin Section 12.4 by proving an existence and uniqueness result for L^p -solutions of (12.1.6). Here G is assumed to be locally Lipschitzian in the L^p -norm. An analogous result on B^∞ -solutions follows. Subsequently, we pursue the investigation of (12.1.7), obtaining a theorem on the existence of L^p -solutions. The proof uses compactness in L^1 . The section is concluded by two results on (12.1.7) in a B^∞ - and L^∞ -setting, respectively.

Section 12.5 is devoted to (12.1.7) in the case where, for each (t, s) , the function $y \mapsto g(t, s, y)$ is locally bounded but not necessarily continuous. The concept of a solution must now be enlarged. A result on the existence of solutions in this enlarged sense is obtained.

Section 12.6 contains two examples. One illustrates the differences between the assumptions of Theorem 12.2.1 and those of Theorem 12.2.8; the other indicates the necessity of one of the assumptions in Theorem 12.2.1.

3.13 Chapter 13: Continuous Dependence, Differentiability, and Uniqueness

Chapter 13 on the *continuous dependence* on parameters, and on the *differentiability and uniqueness* of solutions, is closely related to the previous chapter. First, we consider the simple equation

$$x(\lambda, t) = f(\lambda, t) + \int_0^t a(\lambda, t, s)h(\lambda, s, x(\lambda, s)) \, ds, \quad t > 0. \quad (13.1.6)_\lambda$$

In Theorem 13.2.1 we study the parameter-dependence of the solutions of

$$x(\lambda, t) = G(\lambda, x(\lambda, \cdot))(t), \quad t > 0, \quad (13.1.1)_\lambda$$

under the same hypotheses on G as those that were made in Theorem 12.2.1. Here uniqueness is not assumed. Theorem 13.2.4, related to

Theorem 12.4.1, deals with the parameter-dependence of L^p -solutions of $(13.1.1)_\lambda$. In this case, a Lipschitz condition assures uniqueness.

We analyse the differentiability of the solutions of $(13.1.6)_\lambda$ and $(13.1.1)_\lambda$ with respect to the parameter λ in Section 13.3. The key result of the section is Theorem 13.3.1, where it is shown that under the given hypotheses the solution $x(\lambda, \cdot)$ of $(13.1.1)_\lambda$ has, in a certain sense, a right-hand derivative with respect to λ at $\lambda = 0$. Theorem 13.1.2 is a corollary pertaining to $(13.1.6)_\lambda$.

In Section 13.4, *maximal and minimal solutions* are discussed, and a number of comparison results are given.

The last two sections deal with uniqueness of solutions, in particular with uniqueness of solutions of the equation

$$x(t) = f(t) + \int_0^t g(t, s, x(s)) \, ds, \quad t > 0.$$

A comparison approach together with a monotonicity assumption gives us the uniqueness Theorem 13.5.1. In Theorem 13.5.4 we formulate some results spelling out hypotheses under which the key condition of Theorem 13.5.1 is satisfied.

3.14 Chapter 14: Lyapunov Techniques

In Chapter 14 we study Lyapunov techniques. We stress the main ideas involved, and give some general principles and a few simple examples. As is well known, the actual construction of Lyapunov functions for more complicated equations is frequently an insurmountable task.

After some introductory comments, we discuss, in Section 14.2, the scalar differential-delay equation

$$\begin{cases} x'(t) + a(t)g(x(t)) + b(t)g(x(t-1)) = f(t), & t \geq 0; \\ x(t) = \varphi(t), & -1 \leq t \leq 0. \end{cases}$$

Through the analysis of this equation we are led to two *boundedness principles*. In Proposition 14.2.9, we apply Lyapunov techniques to the scalar nonconvolution integrodifferential equation

$$x'(t) + \int_{[0,t]} \kappa(t, ds)g(x(s)) = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0,$$

and obtain a boundedness result.

In Section 14.3, we examine the existence of limits using Lyapunov methods. Again we emphasize main ideas. We formulate two *limit principles*, and apply them to a differential-delay equation and to a nonlinear Volterra equation. The appendix in Section 14.4 contains some technical lemmas.

3.15 Chapter 15: General Asymptotics

Some of the tools needed for the study of the asymptotics of various nonlinear equations are investigated in Chapter 15. In particular, we study *limit sets*, *limit equations*, and some results in *spectral analysis*. Using these tools and some additional analysis we obtain a large number of asymptotic results in Chapters 17–20.

We begin Section 15.2 by defining the positive and negative limit sets $\Gamma_{\pm}(\varphi)$ of a bounded uniformly continuous function $\varphi : \mathbf{R} \rightarrow \mathbf{C}^n$ as

$$\Gamma_{\pm}(\varphi) = \{ \psi : \mathbf{R} \rightarrow \mathbf{C}^n \mid \varphi(t + t_k) \rightarrow \psi(t) \text{ uniformly on compact subsets of } \mathbf{R} \text{ for some sequence } t_k \rightarrow \pm\infty \},$$

and state some elementary facts about these limit sets. Autonomous operators, the concept of *narrow convergence*, and limit equations are introduced. For the simple equation

$$x(t) + \int_0^t a(t-s)g(x(s)) ds = f(t), \quad t \in \mathbf{R}^+, \quad (15.1.1)$$

where $a \in L^1(\mathbf{R}^+; \mathbf{R}^{n \times n})$ and f vanishes at ∞ , the limit equation is

$$y(t) + \int_{-\infty}^t a(t-s)g(y(s)) ds = 0, \quad t \in \mathbf{R}. \quad (15.1.2)$$

If x satisfies (15.1.1) and $y \in \Gamma_+(x)$, then y satisfies (15.1.2). Some further results in this direction are obtained in Theorems 15.2.9 and 15.2.11. Limit equations associated with periodic equations constitute our final topic in this section.

In Section 15.3, we first give some statements on the limit sets of asymptotically slowly varying functions. We then define finitely generated limit sets and prove that every limit set of this category contains a periodic function.

Section 15.4 characterizes bounded functions in terms of their spectral properties. We begin by introducing the *spectrum* $\sigma(\mu)$ of a finite measure μ (this is the support of the Fourier transform of μ), and continue with a definition of the spectrum of a bounded continuous function φ (this spectrum is defined as $\sigma(\varphi) = \bigcap_{a \in N(\varphi)} \{ \omega \mid \tilde{a}(\omega) = 0 \}$ where $N(\varphi) = \{ a \in L^1(\mathbf{R}; \mathbf{C}) \mid a * \varphi \equiv 0 \}$). In Theorem 15.4.6 we establish some of the properties of these spectra; in particular, we show that $\sigma(\mu * \varphi) \subset \sigma(\mu) \cap \sigma(\varphi)$. Subsequently, a version of Wiener's Tauberian theorem is presented, together with some ramifications.

An important question that frequently arises in asymptotic studies is the following: under what conditions does the inclusion $\sigma(\varphi) \subset \{ \omega \in \mathbf{R} \mid \tilde{\mu}(\omega) = 0 \}$ imply that $\mu * \varphi \equiv 0$? Some results related to this question are given in Theorem 15.4.12, in the following corollaries, and in Theorem 15.4.15. Theorem 15.4.17 characterizes those bounded continuous functions which have (at most) countable spectra.

In Section 15.5, *directional spectra* are defined and some of their properties are proved. Observe that this section is meaningful only when $n > 1$.

Using the *asymptotic (directional) spectrum* we then obtain some asymptotic results. An application to the renewal equation is worked out in Section 15.7. The chapter ends with the proof of a certain Lyapunov-type result which is needed later.

PART III

3.16 Chapter 16: Convolution Kernels of Positive Type

In this part we first develop, in Chapter 16, the theory of convolution kernels of *positive type*. These results are of utmost importance for most of the asymptotic theory derived in later chapters.

A locally finite matrix-valued Borel measure μ supported on $\mathbf{R}^+$ is said to be of positive type if for every $\varphi \in L^2(\mathbf{R}; \mathbf{C}^n)$ with compact support one has

$$[\mu, \varphi] \stackrel{\text{def}}{=} \Re \int_{\mathbf{R}} \left\langle \varphi(t), \int_{\mathbf{R}^+} \mu(ds) \varphi(t-s) \right\rangle dt \geq 0.$$

As Theorem 16.2.5 points out, measures of positive type define tempered distributions. Moreover, μ is of positive type if and only if the real part of the Laplace transform of μ is nonnegative. The equivalence statements are formulated in Theorem 16.2.4 (where an extra size assumption is made on μ) and in the more general Theorem 16.2.6. The section ends with Bochner's Theorem 16.2.7, which characterizes the class of continuous functions of positive type, and with a result on the existence of a weak limit at infinity of a continuous function of positive type.

We give some examples of measures of positive type in Section 16.3. A classic example in the scalar case is $\mu(ds) = a(s)ds$ with a locally integrable, positive, nonincreasing, and convex. Propositions 16.3.5–16.3.7 give some useful results describing how positivity is preserved under multiplication and convolution. The last theorem of this section relates the positivity of a function to the negativity of its second distributional derivative.

In Sections 16.4 and 16.5, certain subclasses of the measures of positive type are investigated in more detail. We first define measures of *strong positive type*: a measure μ is, roughly speaking, of this type if the real part of its Fourier transform is bounded from below by $\epsilon(1 + \omega^2)^{-1}$ for some $\epsilon > 0$. Hypotheses are given under which an absolutely continuous measure $\mu(ds) = a(s)ds$ is of strong positive type. Next, we define measures of *strict positive type*: a measure μ is said to be of this type if there exists a function $a \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ satisfying $\langle y, \Re \tilde{a}(\omega)y \rangle > 0$ for all $y \in \mathbf{C}^n \setminus \{0\}$ and all $\omega \in \mathbf{R}$, such that $\nu(dt) = \mu(dt) - a(t)dt$ is of positive type. We prove that μ is of strict positive type if and only if for all $\tau \in \mathbf{R}$

$$\inf_{|y|=1} \liminf_{\substack{z \rightarrow i\tau \\ \Re z > 0}} \langle y, \Re \hat{\mu}(z)y \rangle > 0.$$

Proposition 16.4.6 maps out the hypotheses under which a nonnegative, nonincreasing, and convex function is of strict positive type.

Section 16.5 is devoted to measures of positive type that satisfy a particular inverse coercivity inequality. Specifically, a locally finite matrix-valued measure μ is of *anti-coercive type*, with coercivity constant $q > 0$, if for every $\varphi \in L^2(\mathbf{R}; \mathbf{C}^n)$ with compact support we have $\mu * \varphi \in L^2(\mathbf{R}; \mathbf{C}^n)$ and

$$\|\mu * \varphi\|_{L^2(\mathbf{R})}^2 \leq q\|\mu, \varphi\|.$$

One immediately sees that if a measure is of anti-coercive type then the operator $\varphi \mapsto \mu * \varphi$ maps $L^2(\mathbf{R}; \mathbf{C}^n)$ continuously into itself.

In Theorem 16.5.6, we characterize measures of anti-coercive type through some equivalence relations. In particular, note that anti-coercivity is equivalent (in the scalar case) to $|\hat{\mu}(z)|^2 \leq q\Re \hat{\mu}(z)$ for $\Re z > 0$. Moreover, we record the important fact that if a kernel of positive type has a resolvent then this resolvent is of anti-coercive type. Some examples of anti-coercive measures are given in Propositions 16.5.7 and 16.5.8.

In Section 16.6 we analyse bounds of the form

$$\left| \int_{\mathbf{R}} \langle b(t), \varphi(t) \rangle dt \right|^2 \leq q\|\mu, \varphi\|,$$

where μ is of positive type, $b \in L^2_{\text{loc}}(\mathbf{R}; \mathbf{C}^n)$, and $\varphi \in L^2(\mathbf{R}; \mathbf{C}^n)$ with compact support. The estimates obtained play a significant role in some of the proofs included in the remaining chapters.

In the first appendix of Chapter 16 we define *semi-inner products*, prove a lemma on these products, and apply it to positive matrices, obtaining some useful inequalities. The second appendix contains some elementary properties of distributions, in particular of *tempered distributions* and of their Fourier and Laplace transforms.

3.17 Chapter 17: Frequency Domain Methods: Basic Results

The analysis concentrates on the two equations

$$x(t) + \int_0^t a(s)g(x(t-s)) ds = f(t), \quad t \in \mathbf{R}^+, \quad (17.1.1)$$

and

$$x'(t) + \int_{[0,t]} \mu(ds)g(x(t-s)) = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0, \quad (17.1.2)$$

where g is a continuous map of $\mathbf{R}^n$ into itself. Occasionally, we allow g to depend explicitly on t . (Chapter 20 contains results on somewhat more complicated equations.)

The methods that we employ in the analysis of (17.1.1) and (17.1.2) are based on the theory of kernels of positive type. Despite the fact that the equations are nonlinear, we make frequent use of transform techniques.

Before discussing Chapter 17 in detail, let us briefly mention the distinctive features of each of Chapters 17–19. Chapter 17 analyses (17.1.1) and (17.1.2) under certain size assumptions on the nonhomogeneous term. For example, to obtain asymptotic statements on the solution x of (17.1.2)

it is typically assumed that $f \in L^1(\mathbf{R}^+; \mathbf{R}^n)$. Chapter 18 takes a somewhat different approach, interpolating between the differentiated and the undifferentiated version of the particular equation under discussion. Finally, in Chapter 19, we allow nonhomogeneous terms f which vanish at infinity, but do not necessarily satisfy integrability assumptions such as $f \in L^p(\mathbf{R}^+; \mathbf{R}^n)$ for $p < \infty$.

Variants of a particular technique are present in several of the proofs of Chapters 17–19. We sketch this technique in the introductory section of Chapter 17. Section 17.2 investigates boundedness of solutions of (17.1.2). Most of these results are formulated in the context of $\mathbf{R}^n$, and require g to be the gradient of a function $G \in C^1(\mathbf{R}^n; \mathbf{R})$. Observe that the conclusion of the various theorems invariably includes

$$\sup_{T>0} Q(\mu, g \circ x, T) < \infty, \quad (3.1)$$

where the quadratic form Q is defined by

$$Q(\mu, g \circ x, T) = \int_0^T \left\langle g(x(t)), \int_{[0,t]} \mu(ds) g(x(t-s)) \right\rangle dt. \quad (3.2)$$

In Section 17.3 we start out from the assumption that x is a bounded solution of (17.1.2) which satisfies (3.1) and then gain a more detailed knowledge about the asymptotic behaviour of x . Lemma 17.3.4 and its consequence Theorem 17.3.5 are important results. They state that, if μ is of strict positive type, $f \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{R}^n)$, g is continuous, and if $x \in BUC(\mathbf{R}^+; \mathbf{R}^n)$ is a locally absolutely continuous solution of (17.1.2) on $\mathbf{R}^+$ satisfying (3.1), then $\lim_{t \rightarrow \infty} g(x(t)) = 0$.

The case where μ is not strictly positive is considered in Theorems 17.3.7 and 17.3.8. It is shown that under the same assumptions as in Theorem 17.3.5, but with strict positivity replaced by mere positivity, one may conclude that the asymptotic spectrum of $g \circ x$ is contained in the set where the real part of the Fourier transform of μ vanishes.

Section 17.4 gives hypotheses under which a solution x of (17.1.2) satisfies $x \in L^2(\mathbf{R}^+; \mathbf{R}^n)$. In Section 17.5 we deduce various asymptotic facts concerning solutions of (17.1.1).

3.18 Chapter 18: Frequency Domain Methods: Additional Results

We continue our frequency domain analysis by presenting some interpolation results. First, in Section 18.2, we interpolate between the integrodifferential and the corresponding integral equation. In Section 18.3 the equation is remoulded by partial integration; in the subsequent section some further methods of recasting the equations are given. Primarily, these methods involve taking the convolution of both sides by a suitable function so that the theory of measures of positive type can be applied. These ideas are applied to integrodifferential equations in Section 18.5.

The miscellaneous assumptions (in terms of Laplace transforms) on the kernel a in

$$x(t) + \int_0^t a(s)g(x(t-s)) \, ds = f(t), \quad t \in \mathbf{R}^+, \quad (18.1.1)$$

and the kernel μ in the integrodifferential equation

$$x'(t) + \int_{[0,t]} \mu(ds)g(x(t-s)) = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0, \quad (18.1.2)$$

are summarized in the introductory section.

For the historical records, we note that the ideas of this chapter were among the first to be exploited in the analysis of nonlinear Volterra equations.

3.19 Chapter 19: Combined Lyapunov and Frequency Domain Methods

Here we consider the equation

$$x'(t) + \int_{[0,t]} \mu(ds)g(x(t-s)) = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0, \quad (19.1.1)$$

under the weakest possible assumptions on the forcing function f . As a consequence, we have to make stronger assumptions on the kernel and the nonlinearity: in Theorem 19.2.1 we require that $\int_{\mathbf{R}^+} t|\mu|(dt) < \infty$, and in Theorem 19.3.1 we require that g be locally Lipschitz-continuous. We assume throughout that there exists a bounded solution. The principal conclusion is that x is asymptotically slowly varying.

In Section 19.4 we use resolvent theory to deal with the case where the kernel μ is not necessarily a finite measure.

3.20 Chapter 20: Monotonicity Methods

Chapter 20 assembles miscellaneous results on the qualitative behaviour of solutions of integral equations. The proofs are based on various kinds of monotonicity techniques.

Section 20.2 is devoted to a brief presentation of *nonconvolution kernels of positive type* under certain convexity assumptions.

The scalar equation

$$x(t) + \int_0^t k(t,s)g(s,x(s)) \, ds = f(t), \quad t \in \mathbf{R}^+, \quad (3.3)$$

is studied in Section 20.3 under the assumption that the kernel k is log-convex, i.e., that it satisfies

$$k(v,s)k(t,u) \leq k(t,s)k(v,u), \quad \text{for almost all } s \leq u \leq v \leq t.$$

In Section 20.4 we define and study kernels of *anti-accretive* (with respect to some L^p -space) and *totally invariant* and *totally positive types*. We give some characterizations of these concepts, and prove results on integral equations with kernels of these categories. In particular, Theorems 20.4.16

and 20.4.17 describe kernels which are, respectively, of L^∞ -anti-accretive type and of totally positive type.

In our final section, we study nonlinear convolution equations of the form

$$x(t) + \int_0^t k(t-s, x(s)) \, ds = f(t), \quad t \in \mathbf{R}^+, \quad (20.5.1)$$

and the corresponding integrodifferential equations

$$x'(t) + \int_0^t k(t-s, x(s)) \, ds = f(t), \quad t \in \mathbf{R}^+; \quad x(0) = x_0. \quad (20.5.2)$$

Note that if, for example, one adds several equations of the form (17.1.2) then one gets an equation of this kind. We prove some asymptotic results on the solutions of these equations using monotonicity assumptions on k in both of its variables.

4. Exercises

1. Rewrite the equation

$$x(t) = \int_S^t h(t, s, x(s)) \, ds + g(t), \quad t \geq S,$$

in the form

$$y(t) = \int_0^t g(t, s, y(s)) \, ds, \quad t \geq 0.$$

2. Show that the temperature $T(0, t)$ in (1.1) satisfies equation (1.2).
3. Rewrite the equation

$$x(t) = h \left(\int_0^t k_1(t, s, x(s)) \, ds, \dots, \int_0^t k_m(t, s, x(s)) \, ds \right) + f(t), \quad t \geq 0,$$

in the form

$$y(t) = \int_0^t g(t, s, y(s)) \, ds, \quad t \geq 0.$$

4. Derive (2.6) from (2.3) and (2.4).
5. Rewrite the equation

$$x(t) = \int_t^T h(t, s, x(s)) \, ds + g(t), \quad t \leq T < \infty,$$

in the form

$$y(t) = \int_0^t g(t, s, y(s)) \, ds, \quad t \geq 0.$$

6. Rewrite the equation

$$\begin{aligned} x'(t) &= - \sum_{i=1}^{\infty} a_i x(t - \tau_i), \quad t \in \mathbf{R}^+, \\ x(t) &= \varphi(t), \quad t \in \mathbf{R}^-, \end{aligned}$$

where each $\tau_i \geq 0$, in the form (1.4). What conditions have to be imposed on the coefficients a_i for the resulting measure μ to satisfy $\mu \in M(\mathbf{R}^+)$?

7. Rewrite the equation

$$x(t) = \int_S^{\alpha(t)} h(t, s, x(\beta(s))) \, ds + g(t), \quad t \geq S,$$

where $\beta(S) = S$, β is strictly increasing and continuously differentiable, and $\beta(\alpha(t)) \leq t$, in the form

$$y(t) = \int_0^t g(t, s, y(s)) \, ds, \quad t \geq 0.$$

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Linear Convolution Integral Equations

*We analyse the equation $x + k * x = f$ ($*$ denotes convolution). We define the resolvent kernel r , and derive the variation of constants formula $x = f - r * f$ for the solution. In addition, we prove the Paley–Wiener theorem, which gives a necessary and sufficient condition for the resolvent to be globally integrable. Some consequences of the Paley–Wiener theorem for the asymptotic behaviour of x are formulated. A short section on convolutions and on Laplace transforms is included.*

1. Introduction

This chapter is devoted to a study of linear Volterra integral equations of the type

$$x(t) + \int_0^t k(t-s)x(s) \, ds = f(t), \quad t \in \mathbf{R}^+. \quad (1.1)$$

Later on we shall consider systems of such equations, but for the time being it suffices to let x , k and f be scalar functions defined on $\mathbf{R}^+$. The functions k and f are known, and we want to find out whether (1.1) has a solution x , and what properties such a solution x may have. We say that k is the *kernel* and that f is the *forcing function*, and we assume that these functions are (at least) locally integrable. Under appropriate additional assumptions on the kernel and on the forcing function, we prove that (1.1) has a unique solution x which behaves roughly in the same way that f does. For example, we give conditions implying that x is bounded on $\mathbf{R}^+$ iff f is bounded on $\mathbf{R}^+$, that x tends to a limit at infinity iff f tends to a limit at infinity, and that x is asymptotically periodic iff f is asymptotically periodic.

First let us show, without going into all the technical details, that similarly to what is true in the theory of linear ordinary differential equations

with constant coefficients, the solution of (1.1) can be expressed by the *variation of constants formula*

$$x(t) = f(t) - \int_0^t r(t-s)f(s) ds, \quad t \in \mathbf{R}^+. \quad (1.2)$$

Here r is the analogue of the fundamental solution of an ordinary differential equation and it is called the *resolvent* of the kernel k .

How does one prove the validity of the formula (1.2)? The idea is very simple, and it consists of setting up an iteration to solve (1.1). We define x_m for $m \geq 0$ by

$$\begin{aligned} x_0(t) &= f(t), \\ x_m(t) &= f(t) - \int_0^t k(t-s)x_{m-1}(s) ds, \quad m \geq 1. \end{aligned} \quad (1.3)$$

This set of equations can be solved recursively for x_m , and one gets an expression for x_m which is an alternating sum of $m+1$ terms, and where each term contains one more integral than the preceding. To be able to deal with these cumbersome expressions we introduce a convenient notation for iterated integrals.

Let us define the *convolution* $a * b$ of two functions a and b , given on $\mathbf{R}^+$, by

$$(a * b)(t) = \int_0^t a(t-s)b(s) ds, \quad t \in \mathbf{R}^+.$$

Then the second equation in (1.3) becomes

$$x_m = f - k * x_{m-1},$$

and the solutions x_m of (1.3) are given by

$$x_m = f - k * f + k * (k * f) - \cdots + (-1)^m \underbrace{(k * (k * \cdots * (k * f) \cdots))}_{m \text{ times}},$$

where the last term contains a total of m convolution factors k and one factor f . We will prove below that if a , b , and c are three locally integrable functions defined on $\mathbf{R}^+$ then $(a * (b * c))(t) = ((a * b) * c)(t)$. In other words, the convolution operator is associative. When we use this fact in the preceding expression for x_m , we obtain

$$x_m = f - r_m * f,$$

where

$$r_m = k - k * k + k * k * k - \cdots + (-1)^{m+1} \underbrace{(k * k * \cdots * k)}_{m \text{ times}}.$$

If we let k^{*j} , $j \geq 1$, denote the $(j-1)$ -fold convolution of k with itself ($k^{*1} = k$), then

$$r_m = \sum_{j=1}^m (-1)^{j-1} k^{*j}.$$

We hope that, as $m \rightarrow \infty$, r_m converges to a limit r equal to

$$r = \sum_{j=1}^{\infty} (-1)^{j-1} k^{*j}, \quad (1.4)$$

and that x_m tends to the solution x of (1.1). If this is so, then passing to the limit in the equation $x_m = f - r_m * f$ we get $x = f - r * f$, which is (1.2).

The function r does not only serve as the resolvent of k in the sense that all solutions can be expressed with the aid of formula (1.2), but r itself satisfies a slightly modified version of (1.1). This can be seen as follows. If we factor out $k*$ to the left (or $*k$ to the right) on the right-hand side of (1.4), then we arrive at

$$r = k - k * r = k - r * k,$$

and we conclude that r is the solution of the two *resolvent equations*

$$r(t) + \int_0^t k(t-s)r(s) \, ds = k(t), \quad t \in \mathbf{R}^+, \quad (1.5)$$

and

$$r(t) + \int_0^t r(t-s)k(s) \, ds = k(t), \quad t \in \mathbf{R}^+. \quad (1.6)$$

In the scalar case it is trivially true that $k*r = r*k$, so these two equations are the same, and they are the particular case of (1.1) where f has been replaced by k . When we deal with systems of equations, then k and r are square matrices, and for matrix-valued functions it is not in general true that $a*b = b*a$. However, the two preceding equations show that even in the system case the kernel k and the resolvent r satisfy $k*r = r*k$. In this case the two equations above are not quite of the form (1.1), because r and k are square matrices, whereas x and f are column vectors.

The preceding iteration scheme is straightforward and gives good results as far as local existence of solutions of (1.1) is concerned. For global results the method has to be substantially refined.

One interesting fact is that it is not important how one constructs a solution r of (1.5) and (1.6). The essential thing is that the solution r of (1.5) and (1.6), if it exists, is unique, and that the existence of a solution r of (1.5) and (1.6) implies that for every appropriately chosen f the equation (1.1) has a unique solution x . Moreover, this solution is given by (1.2). One can see that this is true as follows. Let x satisfy (1.1), and convolve (1.1) by r to get $r*x + r*k*x = r*f$. By (1.6), this implies $k*x = r*f$. Consequently, $x = f - k*x = f - r*f$, and so x is given by (1.2). Conversely, let y be defined by $y = f - r*f$, convolve this relation by k , and use (1.5) to get $k*y = r*f$. Thus $y = f - r*f = f - k*y$, and y satisfies (1.1). (The same argument, with f replaced by k and x replaced by r , can be used to show that the solution r of (1.5) and (1.6) is unique.)

After this sketchy argument we proceed to a more rigorous treatment. First we establish, in Section 3, local existence of the resolvent of k , and

then we continue with global, i.e., asymptotic, considerations in Section 4. Before that, however, we briefly summarize some elementary facts about convolutions and their Laplace transforms.

2. Convolutions and Laplace Transforms

In this section we give a brief description of the convolution of two functions, and of the Laplace and Fourier transforms of functions which are locally or globally integrable. All the proofs are deferred to Section 7. Later on, in Chapters 3, 4 and 16, we shall discuss convolutions where one or both factors are measures or distributions. A brief presentation of the Laplace and Fourier transforms of measures is given in Chapter 3, and Laplace and Fourier transforms of distributions are discussed in Chapter 16.

All of the results in this section are valid both for functions with scalar values in $\mathbf{R}$ or $\mathbf{C}$, and for functions with $(m \times n)$ -matrix values, i.e., with values in $\mathbf{R}^{m \times n}$ or $\mathbf{C}^{m \times n}$. Of course, in all the convolution results we have to require that the dimensions of the matrices should be compatible with respect to matrix multiplication.

2.1 Definition

*The convolution $a * b$ of two functions a and b defined on $\mathbf{R}$ is the function*

$$(a * b)(t) = \int_{-\infty}^{\infty} a(t-s)b(s) \, ds, \quad (2.1)$$

*defined for all $t \in \mathbf{R}$ for which the integral exists. The convolution $a * b$ of two functions a and b defined on an interval $[0, T)$ is the function*

$$(a * b)(t) = \int_0^t a(t-s)b(s) \, ds, \quad (2.2)$$

which is defined for all those $t \in [0, T)$ for which the integral exists.

In the above $0 < T \leq \infty$, and the integrals are interpreted in the sense of Lebesgue; in particular, the integrals are required to converge absolutely. Observe that the convolution on $[0, T)$ can be regarded as a special case of the convolution on $\mathbf{R}$ in the sense that we get (2.2) from (2.1) by defining $a(t) = b(t) = 0$ for $t \notin [0, T)$, applying the convolution on $\mathbf{R}$, and then restricting $a * b$ to $[0, T)$. The case $[0, \infty)$ is especially important.

Occasionally we use a mixture of the two definitions above. For example, if a is defined on $\mathbf{R}^+$ and b is defined on $\mathbf{R}$, then we integrate from $-\infty$ to t .

In the next theorem, and in Theorems 4.5 and 4.6, we let $U \pitchfork V$ stand for the space of functions of the type $\{f + g \mid f \in U, g \in V\}$, where U and V are some function spaces, whose intersection contains nothing but the zero function. Most of the time both U and V will be closed subspaces of some

other space W , and we let the norm in $U \oplus V$ be the same as the norm in W . (If U and V are not closed subspaces of some common space W , then we let the norm of $u + v \in U \oplus V$ be, e.g., $\|u + v\|_{U \oplus V} = \|u\|_U + \|v\|_V$.) In particular, $C(\mathcal{T}_S) \oplus BC_0(\mathbf{R}^+)$ and $AP(\mathbf{R}^+) \oplus BC_0(\mathbf{R}^+)$ can be identified, respectively, with the spaces of continuous, asymptotically periodic, and continuous, asymptotically almost-periodic functions, both with the sup-norm. Similarly, since $L^p(\mathcal{T}_S)$ is the space of S -periodic L^p -functions, and $BL_0^p(\mathbf{R}^+)$ is the space of locally L^p -integrable functions that tend to zero at infinity in an L^p -sense, we find that $L^p(\mathcal{T}_S) \oplus BL_0^p(\mathbf{R}^+)$ is the space of functions that are asymptotically L^p -periodic, endowed with the norm of BL^p .

2.2 Theorem

Let J be $\mathbf{R}$, $\mathbf{R}^+$, or an interval with left end-point 0 and right end-point T , where $0 < T < \infty$.

- (i) Let $a \in L^1(J)$ and $b \in L^p(J)$ for some $p \in [1, \infty]$. Then $a * b \in L^p(J)$, and $\|a * b\|_{L^p(J)} \leq \|a\|_{L^1(J)} \|b\|_{L^p(J)}$. If $p = \infty$, then $a * b \in BUC(J)$, and $\|a * b\|_{\sup(J)} \leq \|a\|_{L^1(J)} \|b\|_{L^\infty(J)}$. If $J = \mathbf{R}$ or $\mathbf{R}^+$, and if $b \in L_\ell^\infty(J)$ then $a * b \in BC_\ell(J)$, and if $b \in L_0^\infty(J)$ then $a * b \in BC_0(J)$. In particular, if b belongs to one of the spaces $B^\infty(J)$, $BC(J)$, or $BUC(J)$, or if $J = \mathbf{R}^+$ or $\mathbf{R}$ and b belongs to one of the spaces $L_\ell^\infty(J)$, $L_0^\infty(J)$, $B_\ell^\infty(J)$, $B_0^\infty(J)$, $BC_\ell(J)$, or $BC_0(J)$, then $a * b$ belongs to the same space as b .
- (ii) Let $J = \mathbf{R}$ or $\mathbf{R}^+$, $a \in L^1(J)$, and $b \in BL^p(J)$ for some $p \in [1, \infty]$. Then $a * b \in BL^p(J)$, and $\|a * b\|_{BL^p(J)} \leq \|a\|_{L^1(J)} \|b\|_{BL^p(J)}$. If $b \in BL_0^p(J)$, then $a * b \in BL_0^p(J)$.
- (iii) Let $a \in L^1(J)$ and $b \in BV(J)$. Then $a * b \in BV(J)$, and $\|a * b\|_{BV(J)} \leq \|a\|_{L^1(J)} \|b\|_{BV(J)}$.
- (iv) Let $J = \mathbf{R}$ or $\mathbf{R}^+$, $a \in L^1(J)$, and $b \in BBV(J)$. Then $a * b \in BBV(J)$, and $\|a * b\|_{BBV(J)} \leq \|a\|_{L^1(J)} \|b\|_{BBV(J)}$. If $b \in BBV_0(J)$, then $a * b \in BBV_0(J)$.
- (v) Let $a \in L^1(\mathbf{R})$. If $S > 0$, and b belongs to one of the spaces $L^p(\mathcal{T}_S)$, $p \in [1, \infty]$, $B^\infty(\mathcal{T}_S)$, $C(\mathcal{T}_S)$, $BV(\mathcal{T}_S)$, or $AP(\mathbf{R})$, then $a * b$ belongs to the same space as b .
- (vi) Let $a \in L^1(J)$. If $J = \mathbf{R}$ or $\mathbf{R}^+$, $S > 0$, and b belongs to one of the spaces $L^p(\mathcal{T}_S) \oplus BL_0^p(J)$, $p \in [1, \infty]$, $B^\infty(\mathcal{T}_S) \oplus B_0^\infty(J)$, $C(\mathcal{T}_S) \oplus BC_0(J)$, $BV(\mathcal{T}_S) \oplus BBV_0(J)$, or $AP(J) \oplus BC_0(J)$, then $a * b$ belongs to the same space as b .
- (vii) Let $J = \mathbf{R}$ or $\mathbf{R}^+$, and let $1 < p, q < \infty$, $1/p + 1/q = 1$, $a \in L^p(J)$ and $b \in L^q(J)$. Then $a * b \in BC_0(J)$ and $\|a * b\|_{\sup(J)} \leq \|a\|_{L^p(J)} \|b\|_{L^q(J)}$.
- (viii) If $a \in L^p(J)$, $b \in L^1(J)$ and $c \in L^q(J)$, for some p and q in $[1, \infty]$ with $1/p + 1/q = 1$, then $((a * b) * c)(t) = (a * (b * c))(t)$, $((b * c) * a)(t) = (b * (c * a))(t)$ and $((c * a) * b)(t) = (c * (a * b))(t)$ for all $t \in J$.
- (ix) If $a, b \in L^1(J)$, and $c \in BL^1(J)$, then $(a * b) * c = a * (b * c)$, $(b * c) * a = b * (c * a)$, and $(c * a) * b = c * (a * b)$.

(x) If a or b is scalar-valued, then $a * b = b * a$.

The analogous results that one gets by replacing $a * b$ in (i)–(vi) by $b * a$ are true as well.

The proof of Theorem 2.2 is given in Section 7.

2.3 Corollary

- (i) Let $a \in L^1_{\text{loc}}(\mathbf{R}^+)$, and $b \in L^p_{\text{loc}}(\mathbf{R}^+)$ for some $p \in [1, \infty]$. Then $a * b \in L^p_{\text{loc}}(\mathbf{R}^+)$. Moreover, if $c \in L^1_{\text{loc}}(\mathbf{R}^+)$, then $(a * b) * c = a * (b * c)$ and $c * (a * b) = (c * a) * b$.
- (ii) Let $a \in L^1_{\text{loc}}(\mathbf{R}^+)$, and $b \in BV_{\text{loc}}(\mathbf{R}^+)$. Then $a * b \in BV_{\text{loc}}(\mathbf{R}^+)$.
- (iii) Let $a \in L^p_{\text{loc}}(\mathbf{R}^+)$ and $b \in L^q_{\text{loc}}(\mathbf{R}^+)$, with $1 \leq p, q \leq \infty$ and $1/p + 1/q = 1$. Then $a * b \in C(\mathbf{R}^+)$ and $(a * b)(0) = 0$. Moreover, if $c \in L^1_{\text{loc}}(\mathbf{R}^+)$, then $((c * a) * b)(t) = (c * (a * b))(t)$ and $((b * c) * a)(t) = (b * (c * a))(t)$ for all $t \geq 0$.

This follows directly from Theorem 2.2 because the restriction of $a * b$ to a finite interval $[0, T)$ depends only on the restrictions of a and b to $[0, T)$.

There is a partial converse to Theorem 2.2(i).

2.4 Theorem

Let J be $\mathbf{R}$, or $\mathbf{R}^+$, or an interval with left end-point 0 and right end-point T , where $0 < T < \infty$, and let a be measurable on J . Then each of the following conditions implies that $a \in L^1(J)$:

- (i) $a * b \in L^\infty(J)$ for every $b \in L^\infty(J)$;
- (ii) $a * b \in L^\infty(J)$ for every $b \in BC_0(J)$;
- (iii) $a * b \in L^1(J)$ for every $b \in L^1(J)$.

The proof is given in Section 7.

On each finite interval $[0, T]$, the convolution operator $b \mapsto a * b$ is compact in $L^p(0, T)$, for all $p \in [1, \infty]$.

2.5 Theorem

Let $0 < T < \infty$, $a \in L^1(0, T)$, and let B be a bounded subset of $L^p(0, T)$, for some $p \in [1, \infty]$. Then the set $\{a * b \mid b \in B\}$ is totally bounded in $L^p(0, T)$.

The proof of this, too, is given in Section 7. (A linear operator is compact iff it maps bounded sets into totally bounded sets, i.e., into sets that can be covered by finitely many balls of radius ϵ , for every $\epsilon > 0$.)

Next, we discuss Laplace and Fourier transforms.

2.6 Definition

The Laplace transform $\hat{a}$ of a function a , defined on $\mathbf{R}^+$, is the function

$$\hat{a}(z) = \int_0^{\infty} e^{-zt} a(t) dt,$$

defined for those $z \in \mathbf{C}$ for which the integral exists as a (absolutely convergent) Lebesgue integral. The (bilateral) Laplace transform of a function a , defined on $\mathbf{R}$, is the function

$$\hat{a}(z) = \int_{-\infty}^{\infty} e^{-zt} a(t) dt,$$

which is defined for those $z \in \mathbf{C}$ for which the integral exists.

Observe that in both cases we have $\hat{a}(z) = (e_z * a)(0)$, where $e_z(t) = e^{zt}$ for $t \in \mathbf{R}$. (In the former case, define $a(t) = 0$ for $t < 0$.) Clearly, if a is matrix-valued, then so is $\hat{a}$.

In the above we have required the Laplace transforms to converge in an absolute sense. In the literature it is quite common to use instead a weaker, Riemann-type convergence condition in the definition of Laplace transforms, but we will not do so here.

In harmonic analysis the *Fourier transform* is used more frequently than the bilateral Laplace transform. They differ from each other only by a rotation in the complex plane; the Fourier transform $\tilde{a}$ of a being defined by $\tilde{a}(z) = \hat{a}(iz)$ for $z \in \mathbf{C}$. Most of the time when we use Fourier transforms we let the argument z be real.

2.7 Theorem

- (i) If $a \in L^1(\mathbf{R})$, then $\hat{a}(z)$ is defined and continuous on the line $\Re z = 0$, $\hat{a}(i\omega) \rightarrow 0$ as $|\omega| \rightarrow \infty$, and $\hat{a}(i\omega) = 0$ for all $\omega \in \mathbf{R}$ if and only if $a(t) = 0$ a.e.
- (ii) If $a \in L^1_{\text{loc}}(\mathbf{R})$, and $\hat{a}(z_0)$ is defined for some $z_0 \in \mathbf{C}$, then $\hat{a}(z)$ is defined on the vertical line $\Re z = \Re z_0$.
- (iii) If $a \in L^1_{\text{loc}}(\mathbf{R})$ and $b \in L^1_{\text{loc}}(\mathbf{R})$, then $(\widehat{a * b})(z) = \hat{a}(z)\hat{b}(z)$ for all those $z \in \mathbf{C}$ for which both $\hat{a}(z)$ and $\hat{b}(z)$ are defined.

The statement in (i) that $\hat{a}(i\omega) \rightarrow 0$ as $|\omega| \rightarrow \infty$ is commonly called the Riemann–Lebesgue lemma.

2.8 Theorem

- (i) If $a \in L^1(\mathbf{R}^+)$, then $\hat{a}(z)$ is defined and continuous in the closed half plane $\Re z \geq 0$, it is analytic in $\Re z > 0$, and $\hat{a}(z) \rightarrow 0$ as $|z| \rightarrow \infty$ in the half plane $\Re z \geq 0$. Moreover, $\hat{a}(z) = 0$ for all z in the half plane $\Re z \geq 0$ if and only if $a(t) = 0$ a.e.
- (ii) If $a \in L^1_{\text{loc}}(\mathbf{R}^+)$, and $\hat{a}(z_0)$ is defined for some $z_0 \in \mathbf{C}$, then $\hat{a}(z)$ is defined in the closed half plane $\Re z \geq \Re z_0$.
- (iii) If $a \in L^1_{\text{loc}}(\mathbf{R}^+)$ and $b \in L^1_{\text{loc}}(\mathbf{R}^+)$, then $(\widehat{a * b})(z) = \hat{a}(z)\hat{b}(z)$ for all those $z \in \mathbf{C}$ for which both $\hat{a}(z)$ and $\hat{b}(z)$ are defined.

Observe that Theorem 2.7 uses bilateral Laplace transforms and Theorem 2.8 standard Laplace transforms. The proofs of these two theorems are given in Section 7.

The following properties of Laplace transforms are easy to verify, so their proofs are left to the reader.

2.9 Theorem

Let a and b be two locally integrable functions defined on $\mathbf{R}$ or $\mathbf{R}^+$.

- (i) If c is a complex constant, $a_c(t) = e^{-ct}a(t)$, and $b_c(t) = e^{-ct}b(t)$, then $(a_c * b_c)(t) = e^{-ct}(a * b)(t)$, and $\hat{a}_c(z) = \hat{a}(z + c)$.
- (ii) If $a_c(t) = a(t/c)$ for some $c > 0$, then $\hat{a}_c(z) = c\hat{a}(cz)$.
- (iii) If a is defined on $\mathbf{R}$, c is real, and $a_c(t) = a(t+c)$, then $\hat{a}_c(z) = e^{cz}\hat{a}(z)$.

3. Local Existence and Uniqueness

Let us return to the Volterra integral equation (1.1). With the convolution notation, this equation can be rewritten in the form

$$x(t) + (k * x)(t) = f(t), \quad t \in \mathbf{R}^+. \quad (3.1)$$

If we suppose that $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{m \times n})$ and that $x \in L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$ for some $p \in [1, \infty]$, then, by Corollary 2.3(i), f belongs to $L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$. The converse question is the interesting one: if $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{m \times n})$ and $f \in L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, then does it follow that (3.1) has a solution $x \in L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$? As we shall see below, the answer to this question is affirmative.

Before we continue, let us agree on what we mean by a solution of (3.1). When all the terms in (3.1) are defined everywhere (e.g., if they are continuous), then we call x a solution of (3.1) on an interval $[0, T]$ or $[0, T)$ if (3.1) holds for all t in this interval. In the L^p -case, where one or several of the terms need not be everywhere defined, we call x a solution of (3.1) on an interval $[0, T)$ if (3.1) holds for almost all $t \in [0, T)$.

To prove that (3.1) has a solution in $L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$ whenever $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{m \times n})$ and $f \in L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, we first show that (3.1) has a 'fundamental' solution r . We give the space $L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ the topology of convergence in the L^1 -norm on each compact subset of $\mathbf{R}^+$.

3.1 Theorem

Let $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$. Then there is a solution $r \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ of the two equations $r + k * r = r + r * k = k$. This solution r is unique and depends continuously on k in the topology of $L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$.

It will be obvious from the proof given below that if $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{R}^{n \times n})$, then $r \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{R}^{n \times n})$.

Proof of Theorem 3.1 First, let us show that a solution $r \in L^1([0, T]; \mathbf{C}^{n \times n})$ of the equations $r + k * r = r + r * k = k$ on an interval $[0, T]$ has to be unique. Suppose that $q \in L^1([0, T]; \mathbf{C}^{n \times n})$ satisfies $q + k * q = k$, and that $r \in L^1([0, T]; \mathbf{C}^{n \times n})$ satisfies $r + r * k = k$. Then

$$r = k - r * k = k - r * (q + k * q) = k - (r + r * k) * q = k - k * q = q.$$

Thus we have uniqueness.

Next, let us observe that it suffices to prove that for each $T \in (0, \infty)$ there is a function $r_T \in L^1([0, T]; \mathbf{C}^{n \times n})$ satisfying

$$r_T + k * r_T = r_T + r_T * k = k$$

on $[0, T]$. If this is true, then by the uniqueness proved above, for each $j \in N$, the restriction of r_{j+1} to $[0, j]$ must be equal to r_j a.e. We get a unique solution $r \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ of $r + k * r = r + r * k = k$ on $\mathbf{R}^+$ by defining $r(t) = r_{j+1}(t)$ for $t \in [j, j+1)$.

In the special case where $\int_0^T |k(t)| dt < 1$, we can use the iteration method outlined in Section 1 to get the function r_T . Define

$$r_m = \sum_{j=1}^m (-1)^{j-1} k^{*j},$$

where k^{*j} is the $(j-1)$ -fold convolution of k by itself. By Theorem 2.2(i),

$$\|k^{*j}\|_{L^1(0, T)} \leq \|k\|_{L^1(0, T)}^j.$$

This means that $\{r_m\}_{m=1}^\infty$ is a Cauchy sequence in $L^1([0, T]; \mathbf{C}^{n \times n})$. Therefore, since $L^1([0, T]; \mathbf{C}^{n \times n})$ is complete, there exists a function $r_T \in L^1([0, T]; \mathbf{C}^{n \times n})$ to which r_m tends in $L^1([0, T]; \mathbf{C}^{n \times n})$ as $m \rightarrow \infty$. Each r_m , $m \geq 2$, satisfies

$$r_m + k * r_{m-1} = r_m + r_{m-1} * k = k.$$

By Theorem 2.2(i), $k * r_{m-1} \rightarrow k * r$ and $r_{m-1} * k \rightarrow r * k$ in $L^1([0, T]; \mathbf{C}^{n \times n})$; hence r must satisfy $r + k * r = r + r * k = k$.

To finish the existence proof, it suffices to show that we may always, without loss of generality, take $\int_0^T |k(t)| dt < 1$. By Lebesgue's dominated convergence theorem, the function $e^{-\sigma t} k(t)$ tends to 0 in $L^1([0, T]; \mathbf{C}^{n \times n})$ as $\sigma \rightarrow \infty$, and so, by choosing σ large enough, we may assume that the function $a(t) = e^{-\sigma t} k(t)$, $t \in [0, T]$, satisfies $\|a\|_{L^1(0, T)} < 1$. Suppose that the equations $q + a * q = q + q * a = a$ have a solution q in $L^1([0, T]; \mathbf{C}^{n \times n})$. Define

$$r(t) = e^{\sigma t} q(t), \quad t \in [0, T].$$

Then $r \in L^1([0, T]; \mathbf{C}^{n \times n})$, and it follows from the definition of the convolution that r satisfies $r + k * r = r + r * k = k$. Thus we have found a solution $r \in L^1([0, T]; \mathbf{C}^{n \times n})$ of the original equation.

We leave the proof of the continuous dependence to the reader. (This claim follows immediately from Lemma 9.3.11.) $\square$

3.2 Definition

The function r in Theorem 3.1 is called the resolvent of k .

In the course of the proof of Theorem 3.1 we make the following two observations that will be useful later.

3.3 Lemma

If $r \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$ is the resolvent of $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$, and σ is a complex constant, then the function $e^{\sigma t}r(t)$ is the resolvent of $e^{\sigma t}k(t)$.

3.4 Lemma

If the function $e^{-\sigma t}k(t)$ belongs to $L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ for some $\sigma \in \mathbf{R}$, then there is a constant c such that the function $e^{-ct}r(t)$ belongs to $L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, where r is the resolvent of k .

In Corollary 4.2 we will get a much sharper version of Lemma 3.4 (the conclusion of Lemma 3.4 is true for every $c \geq \sigma$ such that $\det[I + \hat{k}(z)] \neq 0$ in the half plane $\Re z \geq c$).

Observe that if k is $\mathbf{R}^{n \times n}$ -valued then it follows from the preceding construction that r is $\mathbf{R}^{n \times n}$ -valued, too.

With the resolvent at our disposal we can solve (3.1).

3.5 Theorem

Let $k \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^{n \times n})$. Then, for every $f \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, there is a unique solution $x \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$ of (3.1). This solution is given by the variation of constants formula

$$x(t) = f(t) - (r * f)(t), \quad t \in \mathbf{R}^+, \quad (3.2)$$

where r is the resolvent of k . If V is one of the spaces $L^p_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, with $1 \leq p \leq \infty$, $B^\infty_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, $C(\mathbf{R}^+; \mathbf{C}^n)$, or $BV_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, and if $f \in V$, then $x \in V$.

Proof Let $f \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$ and define x by (3.2). Then $x \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, and

$$x + k * x = x + k * (f - r * f) = x + (k - k * r) * f = x + r * f = f,$$

so x satisfies (3.1).

Conversely, let x be an arbitrary solution of (3.1) in $L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$. Then $f \in L^1_{\text{loc}}(\mathbf{R}^+; \mathbf{C}^n)$, and

$$x = f - k * x = f - (r + r * k) * x = f - r * (x + k * x) = f - r * f,$$

and so x is given by (3.2).

It follows from Corollary 2.3 that x belongs to the same space as f . $\square$

To get an explicit solution of the resolvent equation one can sometimes use Laplace transforms. Suppose that for some sufficiently large σ the function $k(t)e^{-\sigma t}$ belongs to $L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$. Then $\hat{k}(z)$ exists for $\Re z \geq \sigma$, and, by Lemma 3.4, $\hat{r}(z)$ exists in some other half plane $\Re z \geq c$. Using

Theorem 2.8 we conclude that the resolvent equation may be transformed, and that

$$\hat{r}(z) = \hat{k}(z)[I + \hat{k}(z)]^{-1}, \quad \Re z \geq \max\{\sigma, c\}.$$

An inversion of the Laplace transform (which occasionally can be done in a closed form) gives r .

4. Asymptotic Behaviour and the Paley–Wiener Theorems

Suppose that we know something about the asymptotic behaviour of the forcing function f in (3.1). For example, f could be bounded or asymptotically periodic, or f might satisfy $f \in L^p(\mathbf{R}^+; \mathbf{C}^n)$ or $f \in BL^p(\mathbf{R}^+; \mathbf{C}^n)$ for some $p \in [1, \infty]$. What conditions do we need on the kernel k in order to conclude that the solution x of (3.1) has the same asymptotic behaviour as f ?

If $f \in L^1(\mathbf{R}^+; \mathbf{C}^n)$, or $f \in L^\infty(\mathbf{R}^+; \mathbf{C}^n)$, then a necessary and sufficient condition exists, namely $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$. This is true because, by Theorems 2.2(i) and 2.4, the convolution operator $r \ast$ maps $L^1(\mathbf{R}^+; \mathbf{C}^n)$ into $L^1(\mathbf{R}^+; \mathbf{C}^n)$ iff $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, and similarly, it maps $L^\infty(\mathbf{R}^+; \mathbf{C}^n)$ into $L^\infty(\mathbf{R}^+; \mathbf{C}^n)$ iff $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$. In the remaining cases mentioned above, we see, by Theorem 2.2, that the condition $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ is sufficient to imply that x behaves asymptotically in the same way that f does. If k is integrable, then the integrability of r turns out to be a necessary condition as well.

When is it true that $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$? As we shall see below, if $k \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, then one can give a complete answer to this question in terms of a Laplace transform condition. If $k \notin L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, then it is possible to find some necessary conditions, but in this case the problem of finding sharp sufficient conditions is very much open. (We return to this problem in Chapter 6.)

The case where $k \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ is settled by the following theorem.

4.1 The Half Line Paley–Wiener Theorem

Let $k \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$. Then the resolvent r of k satisfies

$$r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$$

if and only if

$$\det[I + \hat{k}(z)] \neq 0, \quad \Re z \geq 0.$$

It is easy to see that the Laplace transform condition is necessary. For, if $r \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$ and $r + k \ast r = k$, then we can transform the equation to get

$$\hat{r}(z) + \hat{k}(z)\hat{r}(z) = \hat{k}(z), \quad \Re z \geq 0$$

(cf. Theorem 2.8). Equivalently,

$$[I + \hat{k}(z)][I - \hat{r}(z)] = I, \quad \Re z \geq 0. \quad (4.1)$$

Clearly this implies that $\det[I + \hat{k}(z)] \neq 0$, $\Re z \geq 0$.

We return later to the more difficult sufficiency part of the proof of Theorem 4.1.

An extension of Theorem 4.1 is given in Section 7.2, where it is assumed that $\det[I + \hat{k}(z)]$ has no zeros on the imaginary axis, but does have zeros in the open half plane $\Re z > 0$. The case where we have zeros on the imaginary axis is treated in Section 7.3.

Theorem 4.1 implies the following sharper version of Lemma 3.4.

4.2 Corollary

If the function $e^{-\sigma t}k(t)$ belongs to $L^1(\mathbf{R}^+; \mathbf{C}^{m \times n})$ for some $\sigma \in \mathbf{R}$, then the function $e^{-ct}r(t)$, where r is the resolvent of k , belongs to $L^1(\mathbf{R}^+; \mathbf{C}^{m \times n})$ for each constant $c \geq \sigma$ such that $\det[I + \hat{k}(z)] \neq 0$ for $\Re z \geq c$.

In the above we assumed that k and r are defined on $\mathbf{R}^+$, and we defined the convolutions $k * r$ and $r * k$ accordingly. In some instances it is useful to study the more general problem where $k \in L^1(\mathbf{R}; \mathbf{C}^{m \times n})$ and $r \in L^1(\mathbf{R}; \mathbf{C}^{n \times m})$, and where k and r satisfy the whole line versions of the resolvent equations, i.e.,

$$r(t) + \int_{-\infty}^{\infty} k(t-s)r(s) ds = r(t) + \int_{-\infty}^{\infty} r(t-s)k(s) ds = k(t), \quad t \in \mathbf{R},$$

or, with our notation for convolutions,

$$r(t) + (k * r)(t) = r(t) + (r * k)(t) = k(t), \quad t \in \mathbf{R}. \quad (4.2)$$

This problem comes up naturally in the course of the proof of Theorem 4.1, and it is of interest in its own right.

The analogue of Theorem 4.1 for equation (4.2) can be stated as follows.

4.3 The Whole Line Paley–Wiener Theorem

Let $k \in L^1(\mathbf{R}; \mathbf{C}^{m \times n})$. Then there is a function

$$r \in L^1(\mathbf{R}; \mathbf{C}^{n \times m})$$

*satisfying the two equations $r + k * r = r + r * k = k$, if and only if*

$$\det[I + \hat{k}(z)] \neq 0, \quad \Re z = 0.$$

This function r is unique.

4.4 Definition

The function r in Theorem 4.3 is called the whole line resolvent of k .

Observe that Theorem 4.3 uses bilateral Laplace transforms as opposed to the standard Laplace transform used in Theorem 4.1. Again, the necessity of the Laplace transform condition is easily proved.

The proof of the sufficiency of the conditions on the determinants in Theorems 4.1 and 4.3 is rather long and somewhat technical. Therefore, let us postpone it for another moment, and show instead how Theorems 4.1 and 4.3 can be applied, respectively, to (3.1), and to the whole line equation

$$x(t) + \int_{-\infty}^{\infty} k(t-s)x(s) ds = f(t), \quad t \in \mathbf{R}.$$

With the convolution notation this equation becomes

$$x(t) + (k * x)(t) = f(t), \quad t \in \mathbf{R}. \quad (4.3)$$

4.5 Theorem

Let $k \in L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$, and let V be one of the following spaces:

- (1) $L^p(\mathbf{R}^+; \mathbf{C}^n)$, $p \in [1, \infty]$, $L^\infty(\mathbf{R}^+; \mathbf{C}^n)$, or $L_0^\infty(\mathbf{R}^+; \mathbf{C}^n)$;
- (2) $B^\infty(\mathbf{R}^+; \mathbf{C}^n)$, $B_\ell^\infty(\mathbf{R}^+; \mathbf{C}^n)$, or $B_0^\infty(\mathbf{R}^+; \mathbf{C}^n)$;
- (3) $BC(\mathbf{R}^+; \mathbf{C}^n)$, $BUC(\mathbf{R}^+; \mathbf{C}^n)$, $BC_\ell(\mathbf{R}^+; \mathbf{C}^n)$, or $BC_0(\mathbf{R}^+; \mathbf{C}^n)$;
- (4) $BL^p(\mathbf{R}^+; \mathbf{C}^n)$, or $BL_0^p(\mathbf{R}^+; \mathbf{C}^n)$, $p \in [1, \infty]$;
- (5) $BV(\mathbf{R}^+; \mathbf{C}^n)$;
- (6) $BBV(\mathbf{R}^+; \mathbf{C}^n)$, or $BBV_0(\mathbf{R}^+; \mathbf{C}^n)$;
- (7) $L^p(\mathcal{T}_S; \mathbf{C}^n) \triangleq BL_0^p(\mathbf{R}^+; \mathbf{C}^n)$, $p \in [1, \infty]$, $B^\infty(\mathcal{T}_S; \mathbf{C}^n) \triangleq B_0^\infty(\mathbf{R}^+; \mathbf{C}^n)$, $C(\mathcal{T}_S; \mathbf{C}^n) \triangleq BC_0(\mathbf{R}^+; \mathbf{C}^n)$, or $BV(\mathcal{T}_S; \mathbf{C}^n) \triangleq BBV_0(\mathbf{R}^+; \mathbf{C}^n)$, where $S > 0$;
- (8) $AP(\mathbf{R}^+; \mathbf{C}^n) \triangleq BC_0(\mathbf{R}^+; \mathbf{C}^n)$.

Then the following four conditions are equivalent:

- (i) the resolvent r of k belongs to $L^1(\mathbf{R}^+; \mathbf{C}^{n \times n})$;
- (ii) $\det[I + \hat{k}(z)] \neq 0$ for all $z \in \mathbf{C}$ with $\Re z \geq 0$;
- (iii) for every $f \in V$, the solution x of the equation $x + k * x = f$ on $\mathbf{R}^+$ belongs to V ;
- (iv) for every $f \in V$, the solution x of the equation $x + k * x = f$ on $\mathbf{R}^+$ belongs to V , and x depends continuously on f (in the norm of V).

That (i) and (ii) are equivalent is exactly the conclusion of the Half Line Paley–Wiener Theorem 4.1. Obviously, (iv) implies (iii). To see that (i) implies (iv) one uses the variation of constants formula (3.2) and Theorem 2.2. The rest of the proof is given in Section 5.

A part of Theorem 4.5 is extended in Section 7.2 to the case where $\det[I + \hat{k}(z)]$ has zeros in the open right half plane, and in Section 7.3 to the case where there are zeros on the imaginary axis.

When we consider equations on the entire real line, the results are practically the same as those for the half line. (There is one exceptional case

that does not appear in Theorems 4.5 and 4.6, namely the case where f is periodic. See Theorem 4.7 below.)

4.6 Theorem

Let $k \in L^1(\mathbf{R}; \mathbf{C}^{n \times n})$, and let V be one of the spaces (1)–(8) in Theorem 4.5, with $\mathbf{R}^+$ replaced by $\mathbf{R}$, or

(9) $AP(\mathbf{R}; \mathbf{C}^n)$.

Then the following four conditions are equivalent:

- (i) k has a whole line resolvent, which belongs to $L^1(\mathbf{R}; \mathbf{C}^{n \times n})$;
- (ii) $\det[I + \hat{k}(z)] \neq 0$ for all $z \in \mathbf{C}$ with $\Re z = 0$;
- (iii) for every $f \in V$, the equation $x + k * x = f$ has a unique solution x in V ;
- (iv) for every $f \in V$, the equation $x + k * x = f$ has a unique solution x in V , and x depends continuously on f (in the norm of V).

When these conditions hold, then the solution x in (iii) and (iv) is given by $x = f - r * f$, where r is the whole line resolvent of k .

Observe that in (iii) and (iv) the solution is required to be *unique* in V ; it does not preclude the possibility that the equation may have a solution that does not belong to any of the spaces V listed above.

The equivalence above of (i) and (ii) is the conclusion of the Whole Line Paley–Wiener Theorem 4.3. To prove the variation of constants formula $x = f - r * f$, one simply repeats the algebraic manipulations in the proof of Theorem 3.5. This formula, together with Theorem 2.2, gives the implication (i) $\Rightarrow$ (iv). Clearly, (iv) implies (iii). The rest of the proof is given in Section 5.

Finally, let us consider the case of periodic functions.

4.7 Theorem

Let $k \in L^1(\mathbf{R}; \mathbf{C}^{n \times n})$, $S > 0$, and let V be one of the following spaces:

- (1) $L^p(\mathcal{T}_S; \mathbf{C}^n)$, $1 \leq p \leq \infty$;
- (2) $B^\infty(\mathcal{T}_S; \mathbf{C}^n)$;
- (3) $C(\mathcal{T}_S; \mathbf{C}^n)$;
- (4) $BV(\mathcal{T}_S; \mathbf{C}^n)$.

Then the following four conditions are equivalent:

- (i) there exists a function $r_S \in L^1(\mathcal{T}_S; \mathbf{C}^{n \times n})$, satisfying

$$\begin{aligned} r_S(t) + \int_0^S k_S(t-s)r_S(s) \, ds &= r_S(t) + \int_0^S r_S(t-s)k_S(s) \, ds \\ &= k_S(t), \quad t \in \mathbf{R}, \end{aligned}$$

where $k_S \in L^1(\mathcal{T}_S; \mathbf{C}^{n \times n})$ is defined by $k_S(t) = \sum_{j=-\infty}^{\infty} k(t+jS)$, $t \in \mathbf{R}$;

- (ii) $\det[I + \hat{k}(i 2\pi m/S)] \neq 0$ for all $m \in \mathbf{Z}$;
- (iii) for every $f \in V$, the equation $x + k * x = f$ has a unique solution x in V ;

(iv) for every $f \in V$, the equation $x + k * x = f$ has a unique solution x in V , and x depends continuously on f (in the norm of V).

When these conditions hold, then the function r_S in (i) is unique, and, for all $f \in V$, the solution x in (iii) and (iv) is given by

$$x(t) = f(t) - \int_0^S r_S(t-s)f(s) ds, \quad t \in \mathbf{R}.$$

Again, (iii) and (iv) do not exclude the possibility that the equation may have solutions that are not S -periodic.

The proof of this, too, is given in Section 5.

4.8 Definition

The function r_S in Theorem 4.7(i) is called the S -periodic resolvent of k .

5. Proofs of the Paley–Wiener Theorems

First, we demonstrate how one deduces Theorem 4.1 from Theorem 4.3.

Proof of Theorem 4.1, assuming Theorem 4.3 The necessity of the Laplace transform condition has already been established. Conversely, suppose that $\det[I + \hat{k}(z)] \neq 0$, for $\Re z \geq 0$. Let $k(t) = 0$ for $t < 0$. By Theorem 4.3, there is a function $r \in L^1(\mathbf{R}; \mathbf{C}^{n \times n})$ satisfying (4.2). The only thing that we have to show is that this function vanishes a.e. on $\mathbf{R}^-$. This we do as follows. Define

$$r_1(t) = \begin{cases} r(t), & t < 0, \\ 0, & t \geq 0, \end{cases} \quad r_2(t) = \begin{cases} 0, & t < 0, \\ r(t), & t \geq 0. \end{cases}$$

Then, for $\Re z = 0$,

$$\hat{r}(z) = \hat{k}(z)[I + \hat{k}(z)]^{-1} = \hat{r}_1(z) + \hat{r}_2(z).$$

Rewrite this as

$$\hat{r}_1(z) = \hat{k}(z)[I + \hat{k}(z)]^{-1} - \hat{r}_2(z), \quad \Re z = 0.$$

Since $\det[I + \hat{k}(z)] \neq 0$ for $\Re z \geq 0$, both $\hat{k}(z)[I + \hat{k}(z)]^{-1}$ and $\hat{r}_2(z)$ are continuous in $\Re z \geq 0$, and analytic in $\Re z > 0$ (cf. Theorem 2.8). Similarly, because we can interpret $\hat{r}_1(-z)$ as an ordinary Laplace transform of the function $r(-t)$, we see that $\hat{r}_1(z)$ is continuous in $\Re z \leq 0$, and analytic in $\Re z < 0$. It follows from, e.g., Rudin [3], Theorem 16.8, (which essentially is a corollary to Morera's theorem) that the function $\varphi(z)$ defined by

$$\varphi(z) = \begin{cases} \hat{r}_1(z), & \Re z \leq 0, \\ \hat{k}(z)[I + \hat{k}(z)]^{-1} - \hat{r}_2(z), & \Re z > 0, \end{cases}$$

is entire. By Theorem 2.8(i), $|\hat{r}_2(z)| \rightarrow 0$ and $|\hat{k}(z)| \rightarrow 0$ as $|z| \rightarrow \infty$ in $\Re z \geq 0$, and this implies that $\varphi(z)$ is bounded in $\Re z \geq 0$. Similarly, $\hat{r}_1(z)$